



Research article

Selective production of light α -olefins and long-chain α -olefins from CO_2/H_2 and CO/H_2 over iron-based catalysts: Effects of Na_2S and H_2O

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ABSTRACT

Conversion of CO_2/H_2 and CO/H_2 into α -olefins via Fischer-Tropsch synthesis (FTS) reaction has stimulated great interest. Herein, iron-based catalysts were developed to produce α -olefins with a tailor-made manner. As-synthesized $1.5\text{Na-Fe}_1\text{Zn}_1$ exhibited high activity for CO_2 conversion, high selectivity toward $\text{C}_2\text{-C}_5$ light olefins along with stable 1000 h run. Separate catalytic cracking of C_{5+} products over ZSM-5 nanosheet promoted light olefins selectivity up to 64.6 %, benefiting from the elimination of hydrogenation reaction. Interestingly, Na_2S showed different effects in CO_2 -FTS and CO -FTS over iron-based catalysts. In detail, Na_2S significantly suppressed catalytic activity and increased selectivity toward alkanes/ CO in CO_2 -FTS reaction. However, it can surprisingly improve catalyst stability in CO -FTS reaction, leading to high activity at about 96 % CO conversion and high selectivity to long-chain C_{4+} olefins at about 65.4 %. In addition, it was demonstrated that H_2O accumulation hindered H_2/CO adsorption and dissociation, subsequent hydrogenation into CH_x , and their carbon-carbon coupling, which resulted in catalyst deactivation and selectivity shifting toward CH_4 and CO in atmospheric pressure CO_2 -FTS reaction.

1. Introduction

The rapidly increasing global energy demand has resulted in a significant reliance on fossil fuels, which gives rise to the substantial emission of greenhouse gas CO_2 from various industrial processes. The emission of CO_2 is reasonably considered to be responsible for various environmental issues such as global warming and ocean acidification. Therefore, it is highly desired to develop efficient technologies that can capture, utilize, and transform CO_2 in order to achieve carbon neutrality [1,2]. Currently, the direct transformation of CO_2 into valuable products such as fuels and chemicals has garnered significant attention [3–17]. Among the products, lower olefins, including ethylene, propylene, and butylene, are high-valued platform molecules for the polymerization industry, premium fuels, solvent, and other chemical intermediates [18–22]. Therefore, the selective conversion of large-scale CO_2 into lower olefins provides an ideal route for addressing the problem of climate change and building the sustainable societies. However, this process has met challenges for its difficulty in the controllable CO_2/CO

adsorption and activation, carbon-chain growth, intermediates desorption to olefins, and good catalyst structure stability.

Up to the present, the direct transformation of CO_2 over iron-based catalysts via Fischer-Tropsch synthesis (FTS) reaction has been reported to be an effective technology route for olefins production [6,16,23]. The iron-based FTS catalysts have demonstrated its high catalytic activity, along with high selectivity to olefins products and low selectivity to undesired CO compared to methanol intermediate-to-olefins (MTO) technology route [24]. It is widely accepted that the CO_2 -FTS process occurs through two-step tandem steps: the reverse water-gas shift (RWGS) reaction over iron oxides (Fe_3O_4), and the subsequent conventional FTS reaction between H_2 and formed CO over iron carbide ($\chi\text{-Fe}_5\text{C}_2$). However, it should be noted that iron-based catalysts without promoters are unfavorable for efficient olefins production, owing to its weak CO_2 adsorption and strong hydrogenation ability to form major light alkanes [25]. The utilization of promoters is widely applied in iron-based CO_2 -FTS process in order to enhance the overall catalytic performance including high reaction activity, targeted

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olefins selectivity, and catalyst stability. In addition, in spite of the capable promotion of targeted olefins by promoters, the olefins distribution is generally following the Anderson-Schulz-Flory (ASF) law, which makes it necessary for us to significantly enhance the selectivity towards light olefins through combined catalytic cracking of higher hydrocarbons over acidic zeolite catalysts and at the same time avoid the generation of undesired methane and other light alkanes.

Alkali metal elements such as Na and K are generally regarded as the electron-donating promoter. The presence of trace alkali metal can bring about higher CO₂ conversion and significant shifting from light alkanes to olefins owing to the promotional adsorption and activation, and the suppressed hydrogenation reaction [16,26]. In spite of the promotional carburization favorable for low concentration CO* intermediate reaction to drive the CO₂-FTS reaction rate, the CO-FTS with high concentration CO in feed gas and alkali metal facilitated CO dissociation shows easier deactivation derived from accumulation of inactive carbon species. It is thus quite difficult to balance CO dissociation to form C* species rate and its subsequent hydrogenation reaction rate in order to achieve stable catalytic activity and product selectivity. However, for CO-FTS reaction, it is favorable for the production of higher-value long-chain α -olefins with high CO conversion owing to the high concentration C* species with high iron carbide (χ -Fe₅C₂) content on catalyst surface.

Besides alkali metal elements, Mn and Zn are frequently applied to be promoter for tuning the catalytic performance. Its effects on reaction process are more complicated owing to its different interactions with iron phases, electron donations, metal oxides (oxygen vacancies) involved reactions, and its structural modification effects, etc [6,16,27–30]. These tunable factors can provide great chances to regulate the reaction performance such as catalytic selectivity and stability. Additionally, sulfur (S) element is generally considered as catalyst poison owing to its strong adsorption with active sites in many reaction processes. However, in iron-based FTS reaction, de Jong et al. have observed the sulfur promoted lower olefins production from syngas with CO-FTS process, which resulted in higher reaction activity, and lower selectivity toward methane [31]. This example suggests that the suitable application of so-called poisons might indicate remarkable positive effects on the concerned issues originating from the suppression of unexpected reaction processes. In the FTS reactions, the sulfur will no doubt affect CO₂/CO adsorption and activation, and thus bring about change in the active phase evolution of iron species, reactant's reaction rate, and accumulation rate of inactive carbon species.

FTS reaction with the hydrogenation of CO produces the hydrocarbon products along with the generation of a large amount of H₂O by removing the dissociative surface O* species at a relatively high pressure. The formed H₂O can interact with both catalyst active phases and oxide support. Especially, in the cobalt-based FTS reaction, owing to the absence of water–gas-shift (WGS) reaction, the formed H₂O could not transform into relatively inert CO₂ with less impact on catalyst structure evolution. In our previous studies on the silica-supported cobalt catalysts, the effects of H₂O on cobalt species evolution and resulted change in catalytic activity and methane selectivity have been intensively investigated [32,33]. In the case of iron-based catalysts for CO-FTS, Ding et al. reported that a hydrophobic FeMn@Si catalyst increases olefins from syngas by reducing CO₂ formation by suppressing the WGS reaction [34].

For CO₂-FTS reaction, the iron-based catalysts generally consist of the iron oxide (Fe₃O₄) and iron carbide (χ -Fe₅C₂) phases for RWGS and CO-FTS reactions, respectively, which together produce about two times of water by-product in contrast to FTS reaction using syngas feedstock. Such a large amount of water inevitably affects catalytic activity and product selectivity by the thermodynamic equilibrium limitation and the influence of iron phases. In our previous studies on CO₂-FTS route for gasoline-range *iso*-paraffins synthesis, we have successfully achieved a substantial promotion in CO₂ conversion level through two-stage reactor system with inter-stage cooling to remove water and introducing non-RWGS active metal component (Co) in the iron catalyst, which is

based on the principle for the release of equilibrium limitation [35,36]. Recently, Zhang et al. designed Fe₅C₂ catalyst covered with a hydrophobic layer. The presence of this hydrophobic layer resulted in the increase in water partial pressure on Fe₅C₂ surface, leading to the reduced CO₂ conversion, along with increasing selectivity to CH₄ [37]. Therefore, it is of great importance to minimize the negative effects of water on the CO₂-FTS reaction.

In this work, based on the above mentioned concerns, we started from the alkali metal (Na) promoted iron catalysts for CO₂-FTS reaction for the production of α -olefins, especially light olefins. The zinc (Zn) element was selected by co-precipitating into iron catalyst, considering its introduction to form ZnFe₂O₄ spinel structure in fresh catalyst and a structure of Fe@ZnO after reduction, which possibly favored catalyst stability during CO₂-FTS reaction. Then, the ZSM-5 nano-sheet was applied for the separate catalytic cracking of collected liquid-phase long-chain products in a separate fixed-bed reactor without the presence of H₂, and thus resulted in the significant increase in selectivity to overall light olefins without obvious effect in the undesired formation of CH₄ and other light alkanes. Furthermore, Na₂S was introduced to regulate the adsorption and activation of both CO₂ and CO, thus investigating its effects on catalyst iron phase evolution, reaction pathways, and catalyst stability over both CO₂-FTS and CO-FTS reactions. Especially, it was expected that the Na₂S added into 1.5Na-Fe1Zn1 catalyst could effectively suppress the carbon deposition rate to improve catalyst stability for the production of higher-value C₄₊ long-chain α -olefins from syngas (CO/H₂). Additionally, a simple hydrophobic strategy was proposed to apply for the CO₂-FTS reaction at both atmospheric pressure and high pressure conditions, which was expected to elucidate major effects of H₂O on catalyst stability and product selectivity over iron-based catalysts for CO₂-FTS reaction through combined catalytic experiments, catalyst characterizations together with theoretical calculations. Generally speaking, we are aiming to develop the high-performance iron-based FeZnNa catalyst with combined stable zeolite acidic catalysis for the selective and stable production of light olefins from CO₂/H₂, and with addition of Na₂S promoter for the selective and stable production of higher-value C₄₊ long-chain α -olefins from CO/H₂.

2. Experimental

2.1. Catalyst preparation.

FeZn catalysts with different Fe/Zn ratios were prepared using the co-precipitation method. Fe(NO₃)₃·9H₂O and Zn(NO₃)₂·6H₂O with total molar amount of 0.18 mol were dissolved into 500 mL of deionized water. The ratios of Fe to Zn were set at 4:1, 2:1, and 1:1, respectively. After stirring at 700 rpm for 30 min, ammonia was slowly added into the solution to achieve a pH of 9. The resulting suspension was further stirred for 1 h and then aged under static conditions for 2 h. After filtration, washing with deionized water and drying at 100 °C for 12 h, dried FeZn powders were obtained. The powders were further calcinated in a muffle furnace at 400 °C for 4 h in air to obtain FeZn catalysts, which were denoted as FexZn1 (x = 1, 2, and 4), where x stand for molar ratios of Fe to Zn. Pure Fe catalyst without Zn addition was prepared using a similar procedure.

Na-FeZn catalysts were prepared through the incipient wetness impregnation of Na₂CO₃ solution into the dried FeZn powders. The Na loading was designated as 0.75 wt%, 1.5 wt% and 3 wt%, respectively. After the impregnation process, the Na-contained catalysts were dried at 100 °C for 12 h, followed by calcination in a muffle furnace at 400 °C for 4 h in air. The resulting catalysts were denoted as yNa-FexZn1 (y = 0.75, 1.5 and 3), where y wt% represents for the Na loading in the catalysts.

Similarly, NaS-FeZn catalysts were prepared using the impregnation of Na₂CO₃ and Na₂S solution into dried FeZn powders (Fe/Zn = 1:1). The total Na loading was 1.5 wt%, while the S loading was 0.1 wt%, 0.2 wt% and 0.3 wt%, respectively. After the impregnation process, the NaS-FeZn catalysts were dried at 100 °C for 12 h, followed by calcination in a

muffle furnace at 400 °C for 4 h in air. The obtained catalysts were denoted as 1.5Na0.1S-Fe1Zn1, 1.5Na0.2S-Fe1Zn1 and 1.5Na0.3S-Fe1Zn1, respectively.

ZSM-5 nanosheet with b-axis thickness of about 30 nm was prepared following the method described in the previous work with minor modification [38]. To obtain seed suspension, 30 g tetraethoxysilane (TEOS), 7.4 g H₂O, and 26.4 g tetrapropylammonium hydroxide (TPAOH) solution (40 wt% in H₂O) were mixed and stirred at 35 °C overnight to fully hydrolyze TEOS. Then, the obtained mixture was dried at 45 °C for about 4 h to remove the formed ethanol from the process of TEOS hydrolysis. Next, it was transferred into a Teflon-lined stainless-steel autoclave and heated at 70 °C for 72 h. To obtain ZSM-5 nanosheet, 3.9 g TEOS, 0.35 g TPABr, 1.07 g TPAOH solution (25 wt% in H₂O), 10.2 g H₂O, and 2.34 g seed suspension were mixed under stirring at 35 °C for 2 h. Then, 0.04 g AlCl₃·6H₂O and 0.56 g NH₄F were added into 8 g H₂O, respectively. The two solutions were slowly added dropwise to the Si-contained mixture under stirring. The resulted mixture was stirred at 35 °C for 2 h, following by being sealed into a Teflon-lined stainless-steel autoclave and aged at 170 °C for 12 h. Finally, ZSM-5 nanosheet was obtained through centrifugation, washing with water for five times, drying at 120 °C overnight, and calcinating at 550 °C for 5 h.

2.2. Catalyst characterization

The qualitative analysis of the phase composition of powder samples was conducted using a Bruker D2 PHASER X-ray diffraction analyzer. Cu Kα (λ = 1.5406 Å) was used as the radiation source. The samples were scanned from 2θ = 5° to 2θ = 90° at a speed of 0.02°/step under the conditions of 30 kV voltage and 10 mA current. N₂ physisorption tests were performed on a Micromeritics ASAP 2020 apparatus at 77 K. Approximately 300 mg of the sample was degassed under vacuum conditions at 200 °C for 4 h to remove moisture and adsorbed impurities. The specific surface area of the samples was calculated using the Brunauer-Emmett-Teller (BET) method, the pore volume was determined based on the adsorption amount at a relative pressure of P/P₀ = 0.9997, and the pore size was calculated using the nitrogen desorption isotherm and the Barrett-Joyner-Halenda (BJH) method. The elemental content of Fe, Zn, Na, and S in the iron-based catalysts was determined using a PerkinElmer Avio 200 Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES). Prior to measurement, the samples were pre-treated by dissolving 0.05 g in nitric acid, diluting to an appropriate concentration, and then subjected to elemental content analysis. Scanning electron microscopy (SEM) of samples were obtained on Hitachi Regulus 8100 field emission scanning electron microscope. Transmission electron microscopy (TEM) testing was performed using the FEI TE20 transmission electron microscope. Elemental distribution of the samples was also analyzed using Super-X energy dispersive X-ray spectroscopy. Thermogravimetric (TG) analysis for spent catalyst was performed on a Mettler Toledo TGA/1100SF thermogravimetric analyzer. The catalyst was heated from 25 °C to 900 °C in O₂ atmosphere at a flow rate of 50 mL/min with a heating rate of 10 °C/min. Mössbauer spectroscopy measurements were conducted using a Wissel spectrometer at with ⁵⁷Co(Pd) as the radiation source. Raman characterization was carried out on a Renishaw inVia spectrometer with a 785 nm excitation.

The H₂ temperature-programmed reduction (H₂-TPR) test of the iron catalyst was conducted using a MicrotracBEL chemisorption analyzer. 100 mg sample was loaded into a quartz tube, followed by Ar treatment at 100 °C for 1 h to remove the moisture. After the temperature dropped below 50 °C, a mixture of 10 % H₂/90 % Ar was introduced into the quartz tube with a controlled flow rate of 30 mL/min. Once the baseline was stable, the temperature was ramped up to 800 °C at a heating rate of 15 °C/min, and the hydrogen signal was recorded using a TCD detector.

Temperature-programmed desorption (TPD) test was performed using the MicrotracBEL chemisorption analyzer. As for H₂-TPD, 100 mg

sample was first reduced in 10 % H₂/90 % Ar atmosphere at 450 °C for 5 h, followed by cooling down to 50 °C in Ar atmosphere. Subsequently, a mixture of 10 % H₂/90 % Ar was introduced into the quartz tube at 50 °C for 30 min, followed by an Ar purge. After the baseline stabilized, the temperature was raised from 50 °C to 850 °C at a rate of 15 °C/min, and the desorption curve was obtained using a mass spectrometer detector. As for CO₂-TPD and CO-TPD, The procedures remained the same, except that the adsorption gases were switched to CO₂ and 10 % CO/90 % Ar, respectively. The subsequent purges and heating atmospheres were in helium. Notably, in the case of TPD test over spent catalysts, similar procedures were used, except that the samples were first treated in Ar atmosphere at 300 °C to remove surface moisture and impurities.

O₂-TPO experiments were performed on the MicrotracBEL chemisorption analyzer. 100 mg sample was loaded into a quartz tube, followed by Ar treatment at 100 °C for 1 h to remove the moisture. After the temperature dropped below 50 °C, a mixture of 10 % O₂/90 % Ar was introduced into the quartz tube with a controlled flow rate of 30 mL/min. Once the baseline was stable, the temperature was ramped up to 900 °C at a heating rate of 15 °C/min. A mass spectrometer was used to detect peak signals at m/z = 28 and 44 in the outlet gas, corresponding to CO and CO₂, respectively.

The contact angle of SiO₂ spheres was measured using the OCA-40 optical contact angle measurement instrument within a contact angle range of 0° to 180°, with an accuracy of ± 0.1°.

Fourier transform infrared (FTIR) spectroscopy of spent catalysts was measured using the Nicolet iS5 infrared spectrometer.

2.3. Catalyst testing

The CO₂ hydrogenation reaction was carried out in a fixed-bed reaction with 1 cm inner diameter. Typically, 2 g catalyst (40–60 mesh) diluted with the same volume of quartz powder (40–60 mesh) was loaded in the reactor. Before the test, the catalyst was reduced in flowing H₂ (40 mL/min) at 450 °C and 0.2 MPa for 5 h. Subsequently, it was cooled down to 290 °C, followed by switching the feed gas (H₂: CO₂: N₂ = 72: 24: 4) into the reaction. Finally, the temperature and pressure were slowly regulated to the desired values (320 °C and 1 MPa) to test the catalytic performance. After reaction, the reaction system was cooled down to room temperature in the feed gas, and then the spent catalyst was in situ passivated in 1 % O₂/99 % N₂ at room temperature for 2 h. The obtained black passivated catalyst was quickly transferred from the reactor to a sample bottle in a continuous-flow argon gas environment. And the sample bottle was fully purged with argon gas and kept with a slight pressure by sealing. Thus, it can avoid the catalyst exposure to air and thereby prevent the oxidation.

The gaseous product was analyzed using an online gas chromatograph (Agilent GC 7820A) equipped with a thermal conductivity detector (TCD) for the analysis of H₂, N₂, CO, CH₄, and CO₂ and a flame ionization detector (FID) for the analysis of hydrocarbons. The condensed liquid products were collected by ice-cooled trap, which were analyzed offline using a gas chromatograph (Shimadzu GC-2014) with a FID. CO₂ conversion (X) and product selectivity (S) were calculated using the following equations, as described in our previous publication [39]:

$$X_{\text{CO}_2} = \frac{[(A_{\text{CO}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}} - (A_{\text{CO}_2})_{\text{out}}/(A_{\text{N}_2})_{\text{out}}]}{(A_{\text{CO}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}}} \times 100\% \quad (1)$$

$$S_{\text{CO}} = \frac{f_{\text{CO}} A_{\text{CO}} / (A_{\text{N}_2})_{\text{out}}}{f_{\text{CO}_2} [(A_{\text{CO}_2})_{\text{in}} / (A_{\text{N}_2})_{\text{in}} - (A_{\text{CO}_2})_{\text{out}} / (A_{\text{N}_2})_{\text{out}}]} \times 100\% \quad (2)$$

$$S_{\text{CH}_4} = \frac{f_{\text{CH}_4} A_{\text{CH}_4} / (A_{\text{N}_2})_{\text{out}}}{f_{\text{CO}_2} [(A_{\text{CO}_2})_{\text{in}} / (A_{\text{N}_2})_{\text{in}} - (A_{\text{CO}_2})_{\text{out}} / (A_{\text{N}_2})_{\text{out}}]} \times 100\% \quad (3)$$

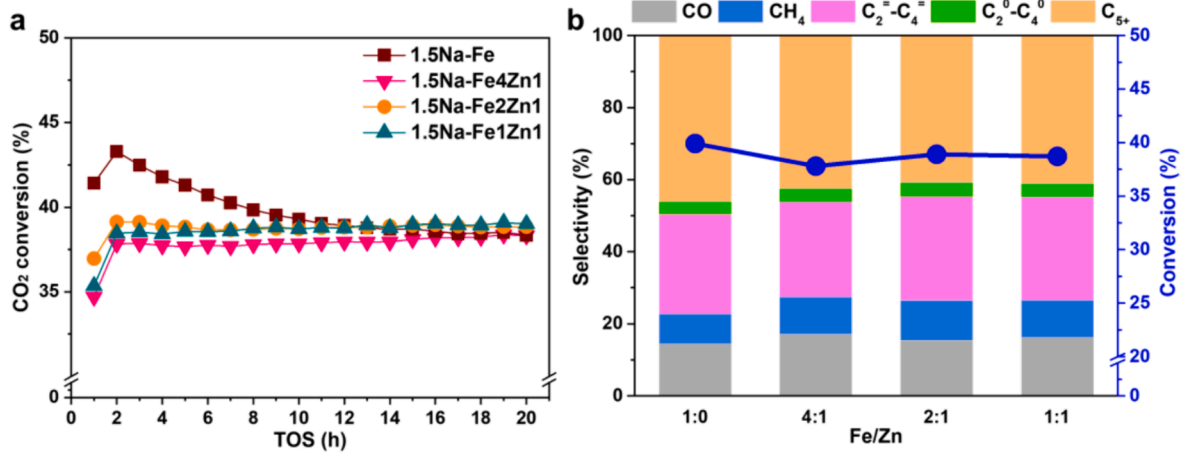


Fig. 1. Catalytic performance of the 1.5Na-FeZn catalysts for CO₂-FTS reaction. Effects of the Fe/Zn ratio on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

$$S_{C_2-C_4} = \frac{S_{CH_4} (A_{C_2^-} + A_{C_3^-} + A_{C_4^-})}{A_{CH_4}} \times 100\% \quad (4)$$

$$S_{C_2^0-C_4^0} = \frac{S_{CH_4} (A_{C_2^0} + A_{C_3^0} + A_{C_4^0})}{A_{CH_4}} \times 100\% \quad (5)$$

$$S_{C_{5+}} = 100 - S_{CH_4} - S_{C_2-C_4} - S_{C_2^0-C_4^0} \quad (6)$$

$$O/(O + P) = \frac{S_{C_2-C_4}}{(S_{C_2-C_4} + S_{C_2^0-C_4^0})} \quad (7)$$

where, A_i represents the peak area of compound i in GC spectrum; in and out represent the feed gas and effluent gas, respectively; f_i is a calibration factor of hydrocarbon compound i and is obtained by external calibration method; $C_2^-C_4^-$, $C_2^0C_4^0$ and C_{5+} represent olefins and alkenes with carbon numbers from 2 to 4 and hydrocarbon products with carbon numbers greater than or equal to 5, respectively; $O/(O + P)$ represents the proportion of olefins in C_2-C_4 hydrocarbon products. The carbon balance over all experimental tests is calculated to be in the range of 96 %–102 %.

The cracking test of the collected liquid-phase products of CO₂-FTS reaction was performed using a U-shaped continuous-flow fixed-bed reactor with 6 mm inner diameter at atmospheric pressure [40]. 0.15 g of zeolite catalyst (40–60 mesh) was loaded in the quartz reactor and pretreated at 550 °C in Ar flow for 1 h. Subsequently, liquid CO₂-FTS products with a speed of 10 μL/min controlled by a pump (LabAlliance Series II) were flowed into an evaporator at 300 °C. The formed vapor was carried out by Ar flow (30 mL/min) and flowed into the quartz reactor. The cracking products were analyzed using online gas chromatograph (GC-14B) equipped with a HP-PLOT Al₂O₃ M capillary column and a FID.

2.4. SSW-NN simulation

In this work, we conducted simulations utilizing global neural network (G-NN) potential, employing the LASP (Large-Scale Atomic Simulation with Neural Network Potential) code available on the website www.lasphub.com [41–43]. The global optimization for the calculation of a multidimensional potential energy surface (PES) and the identification of crucial structures were achieved using stochastic surface walking (SSW) method [44,45]. This approach has been successfully to deal with the complex systems such as zirconia, zeolites and iron carbides [46–48]. The quaternary-element Fe–C–H–O G-NN potential available on the website for LASP: www.lasphub.com/supportings/

FeCHO.pot was obtained through an iterative self-learning process of the SSW-NN data set from various Fe-related configurations, including the bulk and surface of Fe, iron carbides, and various CHO-containing atoms or molecules adsorbed on the surfaces [48]. The ultimate Fe–C–H–O training data set comprises 21,412 structures and is readily accessible on the website: https://www.lasphub.com/supportings/Trainfile_FeCHO.tar.gz. Double-ended surface walking (DESW) method and constrained Broyden dimer (CBD) method were applied for the screening of transition state (TS) for FTS reaction [49,50]. The adsorption energy (E_{ads}) of the adsorbate (X) was calculated using the following equation:

$$E_{ads} = E_{X/slab} - E_{slab} - E_X \quad (8)$$

where, $E_{X/slab}$, E_{slab} , and E_X represented for the total energy of the slab with the adsorbate on it, the slab without any adsorbate on it, and the adsorbate (X) in the gas phase, respectively. Notably, the more negative E_{ads} , the stronger interaction between the slab and the adsorbate.

3. Results and discussion

3.1. Effect of Fe/Zn ratio and Na content on the catalytic performance

Fe-based FTS catalysts with different Fe/Zn ratio and Na content were prepared using co-precipitation and incipient wetness impregnation method. As shown in Fig. 1a, all the catalysts exhibited high CO₂ conversion around 40 %. A similar induction period was observed in the first hour of the reaction, which can be attributed to the evolution of the iron phase in the early stage of the reaction [51]. Upon the introduction of the feed gas, the carburization and oxidation of the iron phase was gradually improved, forming active iron carbide and iron oxide species (especially for Fe₅C₂ and Fe₃O₄) with CO-FTS and RWGS reaction activity. It led to increasing CO consumption, along with the increasing CO₂ conversion. As the evolution of the iron phase slowed down until reaching the phase equilibrium state, the CO₂ conversion and CO selectivity tended to be stable. Notably, among the four catalysts, 1.5Na-Fe without the addition of Zn exhibited a higher initial CO₂ conversion of 43.3 % than that for the Zn-contained catalysts, which might be attributed to the easier carburization of iron without its strong interaction with Zn species. However, as the reaction progressed, this catalyst gradually deactivated, remaining the CO₂ conversion of 38.3 % after 20 h. By contrast, the Zn-contained catalysts showed relatively stable catalytic activity after the induction period of 2 h. It suggested that the addition of Zn significantly improved the stability of catalysts. As shown in Fig. 1b and Table S1, the addition of Zn did not have a significant impact on the lower olefins selectivity. When Fe/Zn ratio was 4, a slight

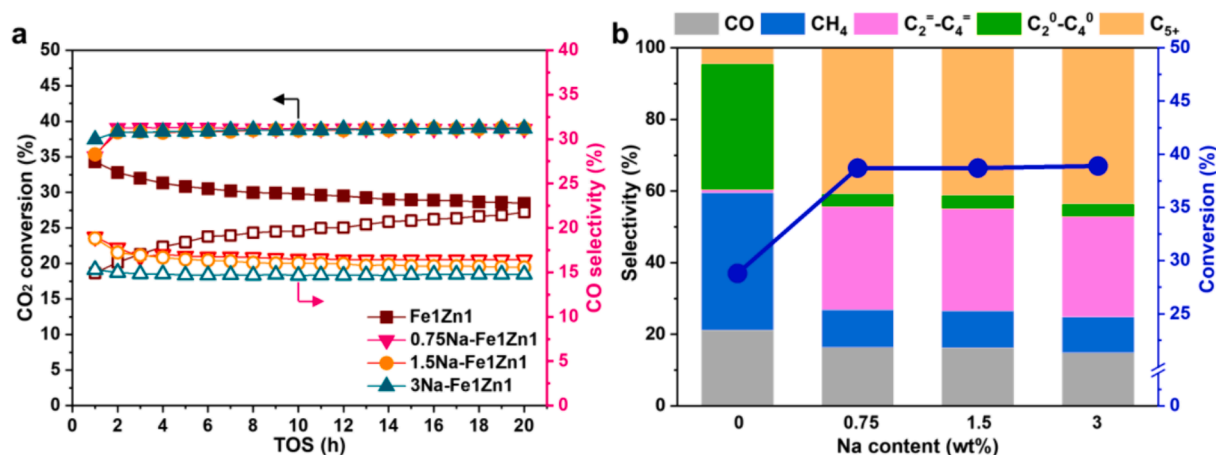


Fig. 2. Catalytic performance of the Na-Fe1Zn1 catalysts for CO₂-FTS reaction. Effects of the Na loading on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

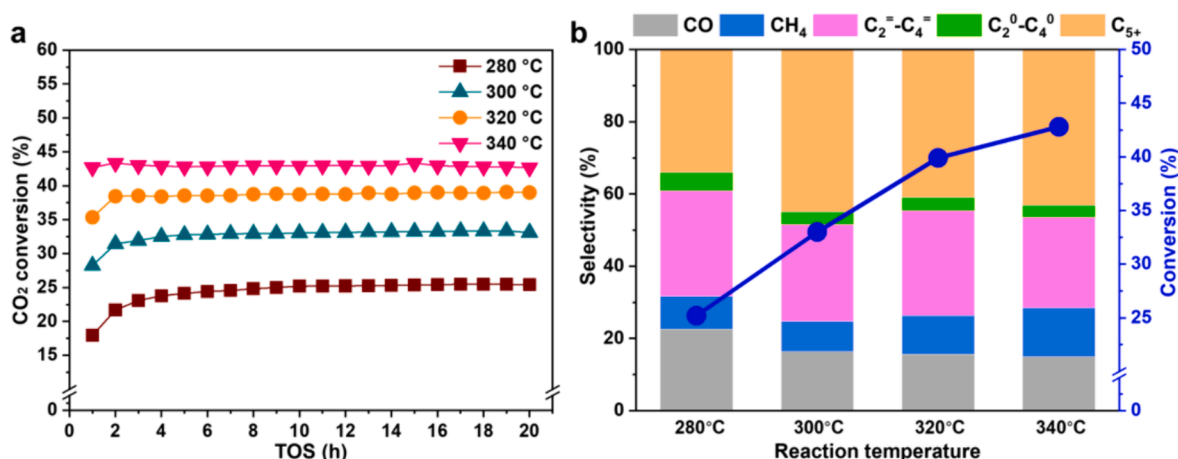


Fig. 3. Catalytic performance of the 1.5Na-Fe1Zn1 catalyst for CO₂-FTS reaction. Effects of the reaction temperature on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

lower CO₂ conversion and lower olefins selectivity were observed. The catalytic performance was optimal when Fe/Zn ratio was 1.

With a fixed Fe/Zn ratio of 1, Na-Fe1Zn1 catalysts with different Na loading were prepared to examine the effect of Na promoter. As shown in Fig. 2 and Table S2, compared to Na-containing catalysts, Fe1Zn1 without Na promoter exhibited evident deactivation accompanied with increasing CO selectivity, which may be attributed to the oxidation of iron carbide (Fe₅C₂) to iron oxide (Fe₃O₄) during CO₂ hydrogenation [52,53]. Meanwhile, the hydrocarbon products primarily consisted of CH₄ and lower alkanes (C₂-C₄), indicating that Na promoter played a critical role in catalyst stability and olefin production. By contrast, the Na-contained catalysts showed evidently higher catalytic activity, reduced selectivity to CO and CH₄, and enhanced production of lower olefins and C₅+. Based on the previous reports, the addition of Na can promote the formation of carbide iron phases by favoring CO dissociation, contributing to the higher catalytic activity and enhanced C-C coupling, and also have a pronounced inhibitory effect on the hydrogenation of hydrocarbon intermediates, facilitating the olefins production [54,55]. When the Na content was 0.75 wt% and 1.5 wt%, the highest lower olefins selectivity was achieved (34.5 %). However, when its content further increased to 3 wt%, lower olefins selectivity slightly decreased to 32.9 % along with a higher C₅+ selectivity of 51.2 %, which is attributed to the enhanced C-C coupling. Based on the results above, it suggested that both Zn and Na promoters were significant for the selective and stable conversion of CO₂ into lower olefins.

3.2. Effect of reaction conditions on the catalytic performance

The optimal 1.5Na-Fe1Zn1 catalyst was further tested under various reaction conditions to improve the catalytic performance for lower olefins production. The effect of reaction temperature was firstly examined. As shown in Fig. 3 and Table S3, with an increase in the reaction temperature from 280 °C to 340 °C, CO₂ conversion increased from 25.2 % to 42.8 %, accompanied by the decline in CO selectivity from 22.6 % to 14.9 %. Meanwhile, the 1.5Na-Fe1Zn1 catalyst exhibited excellent catalytic stability at all four temperatures. Furthermore, it could be observed that the higher reaction temperature was conducive to reducing the induction period, which is related to the evolution of the iron phase in the early stages of the CO₂-FTS reaction. The higher temperature favored the activation and dissociation of reactant molecules, which could promote the formation of active iron carbide phases and reduce the induction period. The STY (space time yield) values of lower olefins were calculated to obtain the proper reaction temperature for lower olefins production (Figure S1). When reaction temperature was 320 °C, the highest lower olefins STY was calculated to be 20.4 mg·g_{cat}⁻¹·h⁻¹. Obviously, when the temperature was below 320 °C, the lower CO₂ conversion and higher CO selectivity contributed to the decline in lower olefins production from CO₂-FTS reaction. When the temperature increased to 340 °C, more CH₄ and C₅+ were generated, leading to the decreased lower olefins production.

As shown in Fig. 4 and Table S4, CO₂ conversion on the 1.5Na-

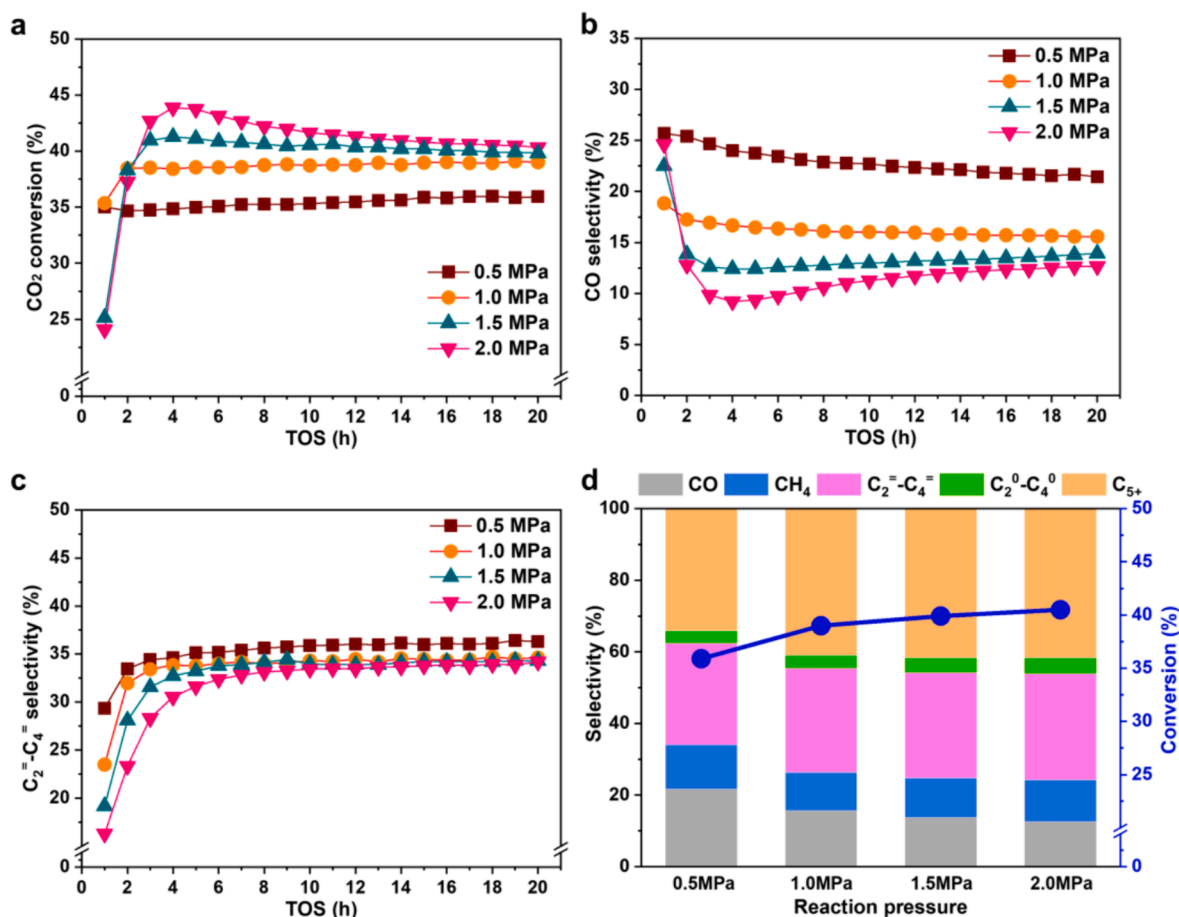


Fig. 4. Catalytic performance of the 1.5Na-Fe1Zn1 catalyst for CO₂-FTS reaction. Effects of the reaction pressure on the (a) catalytic activity, the selectivity of (b) CO and (c) lower olefins, and (d) product distribution. Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

Fe1Zn1 catalyst increased with an increase in the reaction pressure, along with the decreased CO selectivity. Since CO-FTS reaction is volume contracting, the higher reaction pressure reasonably facilitates CO transformation. It contributed to the enhanced CO consumption via FTS reaction over Hagg iron carbide phase, leading to the decline in CO selectivity. On the other hand, the consumption of the CO product from the RWGS reaction over magnetite phase further promoted CO₂ conversion by equilibrium shifting. Interestingly, the higher reaction

pressure resulted in a lower initial activity, longer induction period, and more evident decline in CO selectivity in the initial stage of reaction. It might be related to the enhanced competition adsorption of H₂ species on catalyst surface at higher pressure, which brought about the decline in adsorbed surface CO* species and promoted the hydrogenation of C* species derived from CO dissociation [56]. Consequently, the insufficient concentration of surface C* species slowed down the formation of active iron carbide phases, leading to a lower initial activity

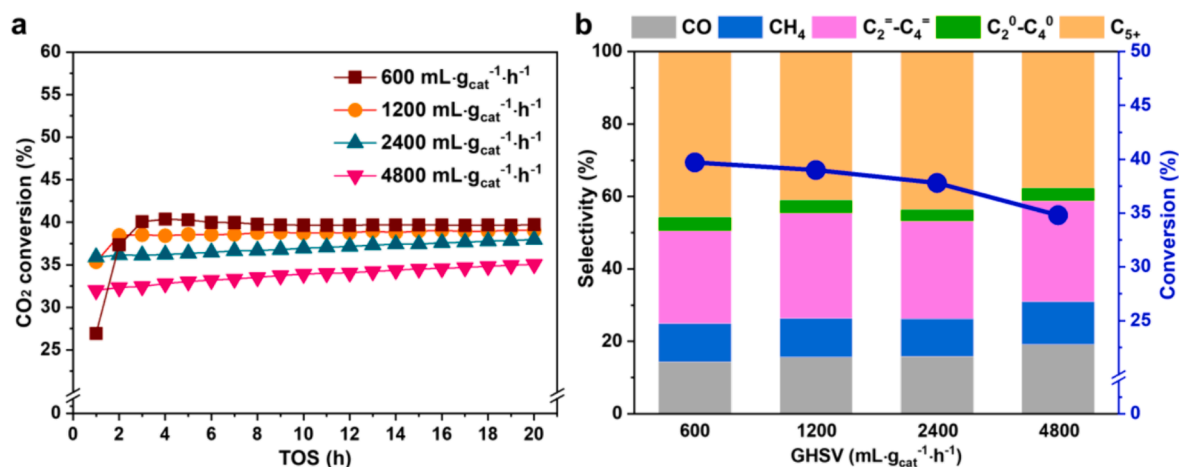


Fig. 5. Catalytic performance of the 1.5Na-Fe1Zn1 catalyst for CO₂-FTS reaction. Effects of the GHSV on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, 320 °C, 1 MPa, and H₂/CO₂ = 3.

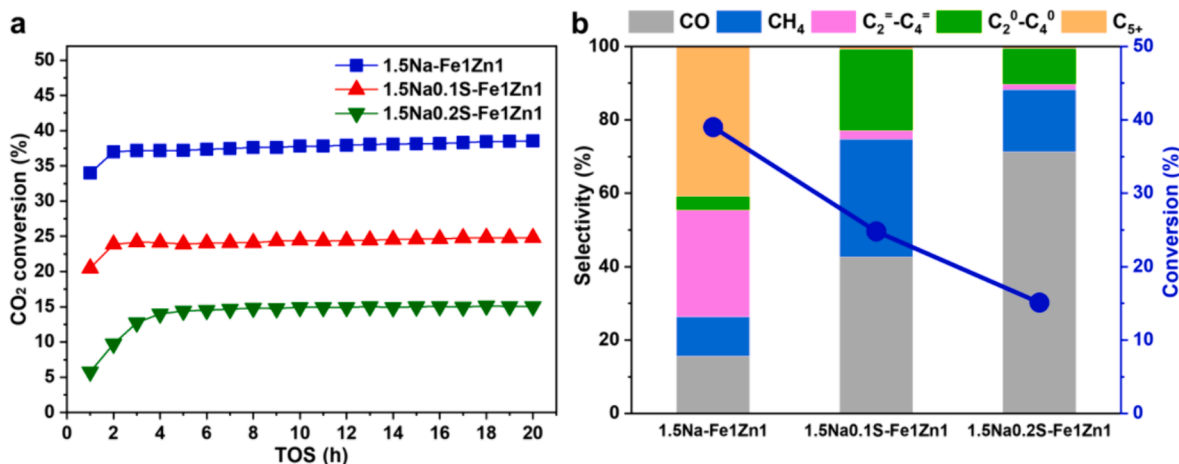


Fig. 6. Catalytic performance of the NaS-Fe1Zn1 catalyst for CO₂-FTS reaction. Effects of the S content on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2.0 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

and longer induction period. At higher reaction pressure (2 MPa), the selectivity to CO slowly increased with the decline in catalytic activity with reaction time on stream, which was ascribed to the higher pressure with higher steam concentration to be more favorable for the oxidation of surface iron carbide. A similar phenomenon was observed in other report [51]. And when the oxidation-carburization tended to be equilibrium state, the catalytic activity and selectivity to CO became stable. Moreover, as shown in Table S4, with the increase of reaction pressure from 0.5 MPa to 2 MPa, lower olefins selectivity slightly decreased from 36.2 % to 33.9 %, accompanied by the decrease of O/(O + P) from 89.4 % to 87.2 % and higher selectivity to methane and C₂-C₄ alkanes. It suggested that high-pressure conditions were more favorable for the hydrogenation of surface hydrocarbon intermediates.

As shown in Fig. 5 and Table S5, 1.5Na-Fe1Zn1 catalyst was further tested at different gas hourly space velocity (GHSV). With the increase of GHSV from 600 mL·g_{cat}⁻¹·h⁻¹ to 4800 mL·g_{cat}⁻¹·h⁻¹, CO₂ conversion decreased from 39.7 % to 34.8 %, along with the increasing CO selectivity from 14.3 % to 19.1 %. As shown in Figure S2a and b, according to thermodynamic analysis on RWGS reaction, the equilibrium constant at 320 °C was calculated to be about 0.033, along with CO₂ equilibrium conversion at about 25.8 %. It suggested that the hydrogenation of CO into hydrocarbons could promote CO₂ conversion in CO₂-FTS reaction (Figure S2c). The equilibrium degree of RWGS reaction was further calculated using the following equation and shown in Figure S2d, denoted as $\eta(\text{RWGS})$.

$$\eta(\text{RWGS}) = \frac{(P_{\text{H}_2\text{O}}/P_{\text{total}}) \times (P_{\text{CO}}/P_{\text{total}})}{(P_{\text{CO}_2}/P_{\text{total}}) \times (P_{\text{H}_2}/P_{\text{total}})} \times \frac{1}{K} = \frac{(F_{\text{H}_2\text{O}})_{\text{out}} \times (F_{\text{CO}})_{\text{out}}}{(F_{\text{CO}_2})_{\text{out}} \times (F_{\text{H}_2})_{\text{out}}} \times \frac{1}{K} \quad (9)$$

where, P_i (Pa) represented for the partial pressure of product or reactant i ($i = \text{CO}, \text{H}_2\text{O}, \text{CO}_2$ and H_2), P_{total} represented for the total pressure, $(F_i)_{\text{out}}$ (mol/min) represented for the flow rate of i out of the reactor, and K represented for the equilibrium constant at reaction temperature.

It was found that when GHSV was at 600 mL·g_{cat}⁻¹·h⁻¹ and 1200 mL·g_{cat}⁻¹·h⁻¹, $\eta(\text{RWGS})$ was calculated to be 1.0, suggesting the RWGS reaction was at equilibrium under a lower GHSV. When GHSV further increased to 2400 mL·g_{cat}⁻¹·h⁻¹ and 4800 mL·g_{cat}⁻¹·h⁻¹, $\eta(\text{RWGS})$ gradually decreased to 0.9 and 0.8, respectively.

In CO₂-FTS reaction, the higher gas velocity would reduce the contact time between primary product CO and FTS catalyst, resulting in a higher CO selectivity. In the case of CO₂ conversion, it could be affected by reaction thermodynamics and kinetics. As shown in Figure S3, it was found that the equilibrium conversion of CO₂ increased along with the decline in CO selectivity. Therefore, in present work, a lower CO selectivity under low space time contributed to a high CO₂ equilibrium

conversion, which was the reason for the higher CO₂ conversion under the GHSV of 600 mL·g_{cat}⁻¹·h⁻¹ and 1200 mL·g_{cat}⁻¹·h⁻¹. In contrast, when the GHSV increased up to 2400 mL·g_{cat}⁻¹·h⁻¹ and 4800 mL·g_{cat}⁻¹·h⁻¹, the resulted higher CO selectivity decreased CO₂ equilibrium conversion. Moreover, the shorter contact time between CO₂/H₂ and catalyst couldn't enable the equilibrium of RWGS reaction, leading to more rapid decline in CO₂ conversion. As for hydrocarbon product distribution, with the increase in GHSV, especially for 4800 mL·g_{cat}⁻¹·h⁻¹, the increasing CH₄ selectivity and decreasing C₅₊ selectivity were observed. Generally speaking, a relatively lower GHSV of 1200 mL·g_{cat}⁻¹·h⁻¹ was favorable for the production of lower olefins along with suppressing selectivity to CO and methane.

As shown in Figure S4 and Table S6, the effect of H₂/CO₂ ratio was further examined. With the increase of H₂/CO₂ ratio from 1 to 3, CO₂ conversion increased from 17.5 % to 39.0 %, along with the decline in CO selectivity from 30.5 % to 15.6 %. It was attributed to the enhanced supply of H* in CO₂ and CO hydrogenation. As for hydrocarbon product distribution, the enhanced supply of H* in CO₂-FTS also facilitated the termination of carbon chain growth, leading to evidently enhanced lower olefins selectivity from 29.5 % to 34.5 % along with a slight increase in CH₄ selectivity from 11.6 % to 12.7 %. Therefore, a higher H₂/CO₂ ratio of 3 was favorable for lower olefins production, owing to the significantly enhanced CO₂ conversion and lower olefins selectivity along with suppressed selectivity to CO.

Based on the results of reaction conditions screening, it can be concluded that in the CO₂-FTS reaction, high temperature, high pressure, low GHSV, and high H₂/CO₂ ratio conditions can enhance CO₂ conversion and hydrocarbon product selectivity. However, in the case of lower olefins production, the excessively high pressure and high temperature and low GHSV did not favor lower olefins selectivity. Taking all factors into account, the optimal reaction conditions for the production of lower olefins over iron-based CO₂-FTS were selected under 320 °C, 1 MPa, 1200 mL·g_{cat}⁻¹·h⁻¹, and H₂/CO₂ = 3.

3.3. Effect of Na₂S on the catalytic performance

Under the optimized reaction conditions, the influence of Na₂S on the performance of 1.5Na-Fe1Zn1 catalyst in the CO₂-FTS reaction was investigated. Keeping the total Na content constant, the addition of Na₂S was carried out by replacing part of Na₂CO₃ with Na₂S in the impregnation process. As shown in Fig. 6 and Table S7, the addition of S significantly reduced catalyst activity, accompanied by a sharp increase in CO selectivity. Correspondingly, after the addition of Na₂S, CH₄ and lower alkanes were the major hydrocarbon products, while olefins were very limited. It suggested that Na₂S could poison the active sites of the

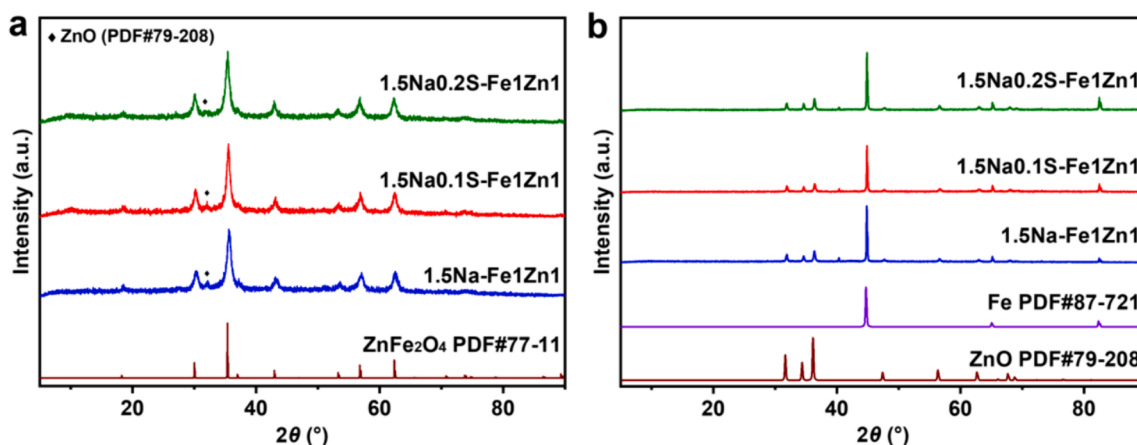


Fig. 7. XRD patterns of the (a) fresh and (b) reduced catalysts.

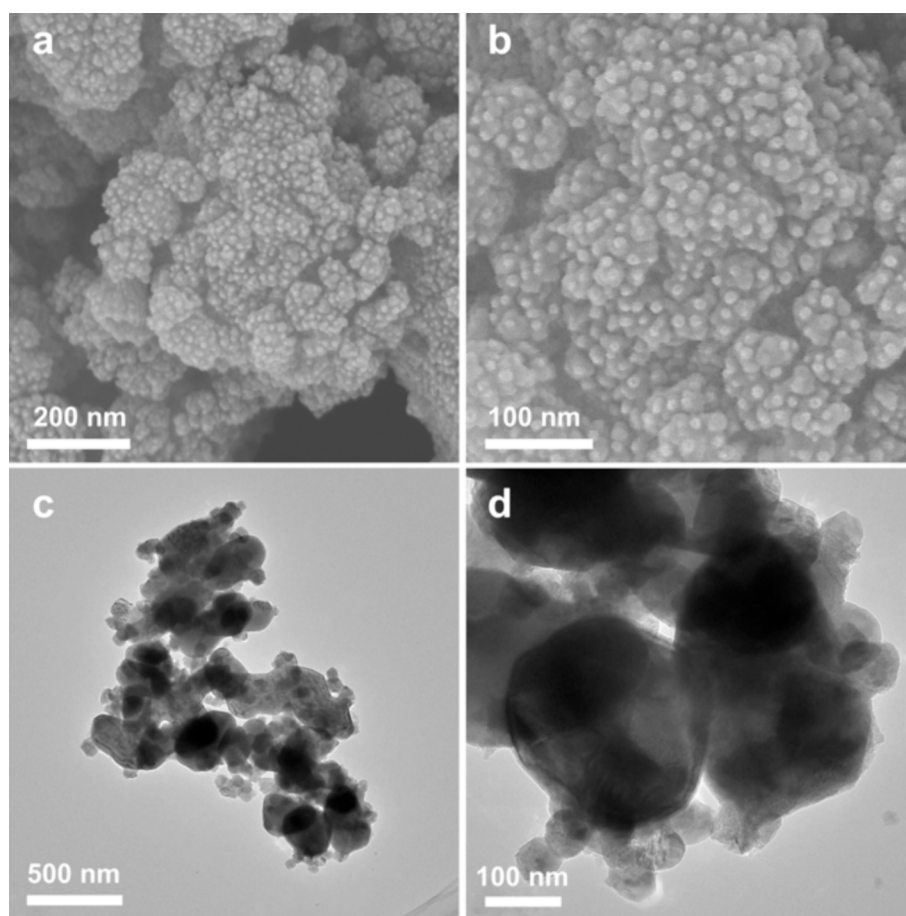


Fig. 8. (a, b)SEM images of the fresh 1.5Na-Fe1Zn1 catalyst, and (c, d) TEM images of the reduced 1.5Na-Fe1Zn1 catalyst.

FTS reaction, enhance the hydrogenation of surface hydrocarbon intermediates, and inhibit C–C coupling. Additionally, as the Na_2S content increased, catalyst induction period significantly increased, indicating that the addition of Na_2S inhibited the evolution of the iron phase at the early stage of reaction.

3.4. Characterizations of 1.5Na-Fe1Zn1 and NaS-Fe1Zn1 catalysts

3.4.1. Characterizations of fresh and reduced catalysts

The XRD patterns of fresh and reduced catalysts are presented in Fig. 7. All the fresh catalysts exhibited ZnFe_2O_4 (PDF#77-11) phase.

Meanwhile, the weak peak at 32° could be corresponding to ZnO (PDF#79-208). No Na and S related peak was detected. After reduction, ZnFe_2O_4 was transformed into metallic Fe^0 (PDF#87-721) and ZnO . The higher peak intensity for Fe^0 suggested a better crystallinity of Fe^0 than that of ZnO . In addition, the XRD patterns of reduced S-contained catalysts were similar to that of 1.5Na-Fe1Zn1, suggesting that the addition of small amount of S did not affect catalyst phase after reduction. Based on BET and ICP-OES, textural properties and promoter content of fresh catalysts are listed in Table S8. After the addition of Zn, catalyst surface area increased from $37 \text{ m}^2/\text{g}$ to $47 \text{ m}^2/\text{g}$, indicating that Zn promoter could enhance the dispersion of catalysts. Meanwhile, the addition of S

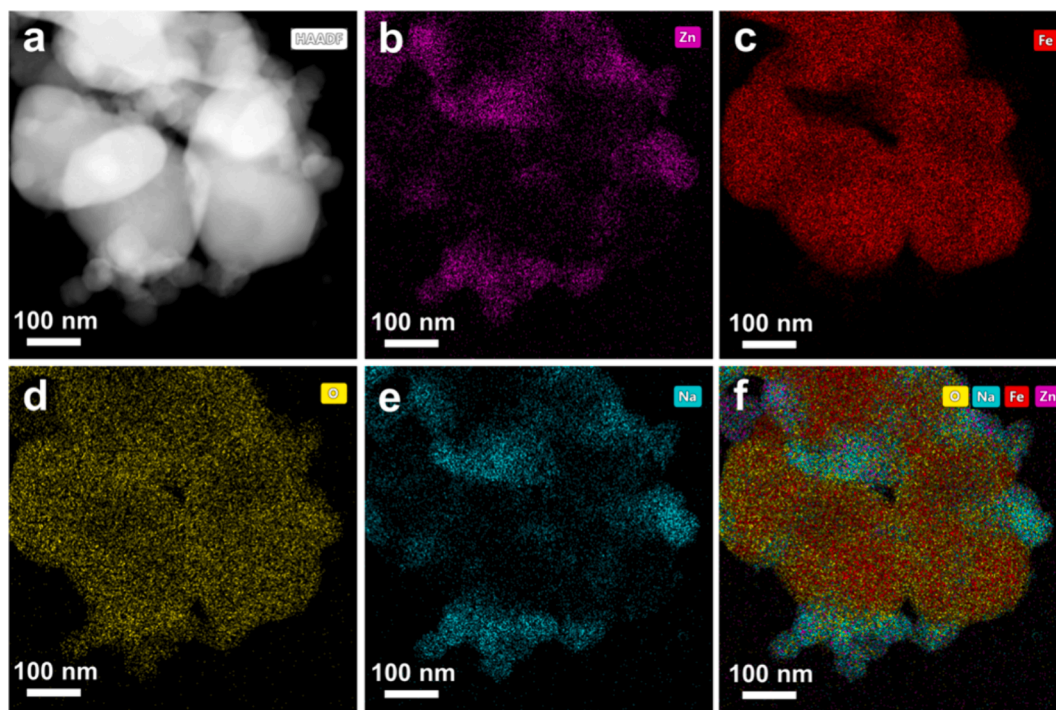


Fig. 9. (a) HADDF-STEM image and (b-f) EDX mapping of the reduced 1.5Na-Fe1Zn1 catalyst.

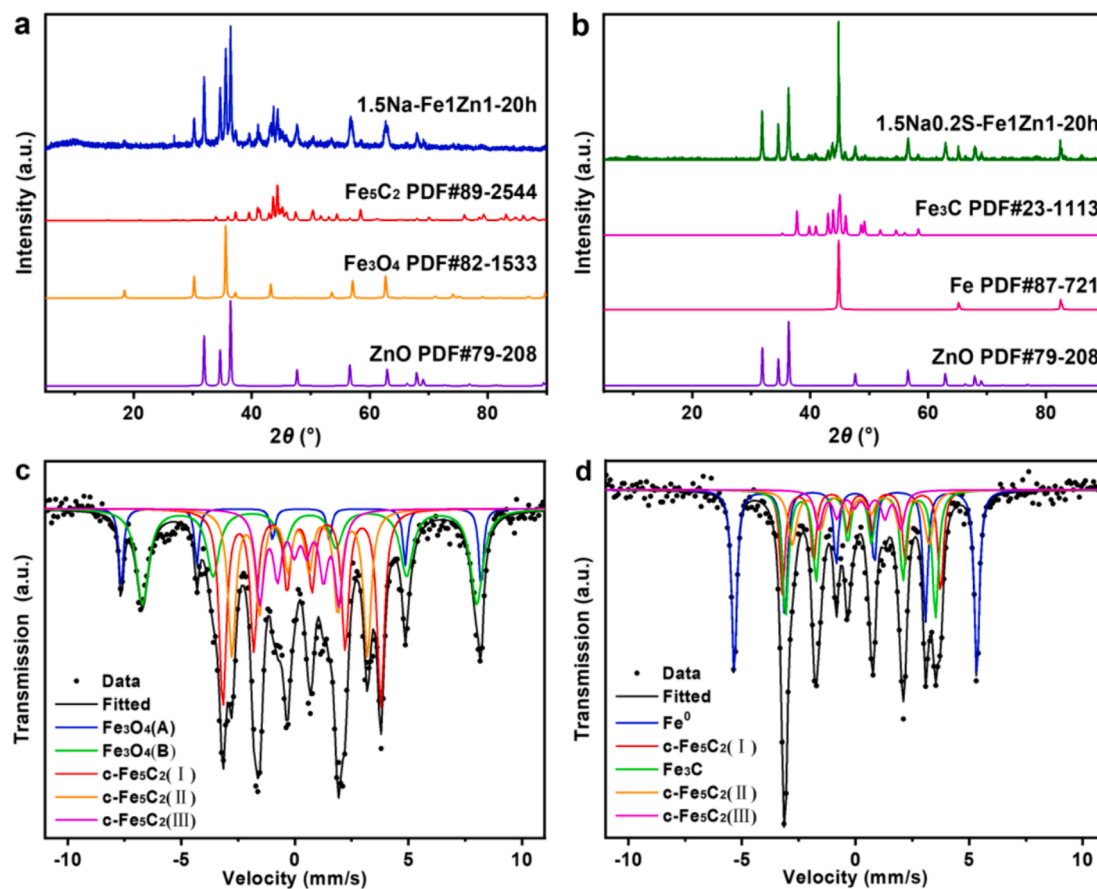


Fig. 10. XRD patterns of spent (a) 1.5Na-Fe1Zn1 and (b) 1.5Na0.2S-Fe1Zn1 catalyst, and Mössbauer spectroscopy of spent (c) 1.5Na-Fe1Zn1 and (d) 1.5Na0.2S-Fe1Zn1 catalyst for CO_2 -FTS.

did not show apparent effect on catalyst textural properties. According to the ICP-OES result, the content of Fe and Zn in fresh 1.5Na-Fe1Zn1 was 44.5 wt% and 25.5 wt%, respectively. The lower Zn content than that for Fe should be related to its acid-alkaline amphoteric properties to be partially dissolved in the precipitation process. The actual content of promoters was approximately similar to the theoretical content. The Na content in all fresh catalysts was in the range of 1.7–1.8 wt%. Meanwhile, the S content in 1.5Na0.1S-Fe1Zn1 and 1.5Na0.2S-Fe1Zn1 was 0.11 wt% and 0.22 wt%, respectively.

To check the effect of Na₂S addition on catalyst reduction behavior, H₂-TPR was performed. As shown in Figure S5, three reduction peaks could be recognized, corresponding to the transformation from ZnFe₂O₄ to Fe₃O₄ at about 335 °C, to FeO at about 425 °C, and then to Fe at about 610 °C. With the increase of Na₂S content, the reduction peak slightly shifted to a lower temperature, indicating that Na₂S addition favored the reduction of catalyst. It should be related to the electron withdrawing effect of Na₂S. The addition of Na promoter into metal oxides usually has a negative effect on catalyst reduction, which may be attributed to the enhanced stabilization of H⁺ species from H₂ heterolytic dissociation through electron donating, and the subsequent suppression of the electron donation from H⁺ species to metal cation for water formation [57]. By contrast, the addition of small amount of Na₂S would weaken the electron donating effect of Na promoter and enhance catalyst reduction.

The SEM images of the fresh 1.5Na-Fe1Zn1 are presented in Fig. 8a and b. The catalyst exhibited a block-like particle morphology with dense protrusion structure distributed on its surface. Combined with the XRD results in Fig. 7a, it can be inferred that the bulk structures should be corresponding to the ZnFe₂O₄ spinel nanoparticles, while these protrusions consisted of smaller ZnO nanoparticles with relatively poor crystallinity. In our previous study, it was found that non-supported iron-based catalysts underwent significant structural changes during the reduction process [4]. The reduction process would result in the aggregation of iron particles due to the remove of oxygen atoms from iron oxides. Therefore, the addition of structure promoters such as Mn, and Zn elements was very significant for promoting the dispersion of Fe species. The TEM images of the reduced 1.5Na-Fe1Zn1 catalyst are shown in Fig. 8c and d. Clearly, after reduction, 1.5Na-Fe1Zn1 catalyst consisted of both relatively large spherical nanoparticles with diameters ranging from 260 to 290 nm and smaller nanoparticles with diameters ranging from 60 to 70 nm attached to their surfaces. Combined with the corresponding XRD results in Fig. 7b, it can be inferred that the larger particles were composed of metallic Fe⁰, while the smaller particles should be assigned to ZnO.

To obtain the element distribution of the reduced catalyst, HAADF-STEM and EDX mapping were conducted. The corresponding images of reduced 1.5Na-Fe1Zn1 are presented in Fig. 9. Larger spherical Fe nanoparticles with smaller nano-sized ZnO on the surface of Fe nanoparticles could be clearly observed, suggesting the formation of Fe@ZnO structure after the reduction process. It was consistent with the TEM images. Furthermore, it was observed that the distribution regions of Na and Zn elements highly overlapped. It might be related to the abundance of hydroxyl groups on the ZnO surface, which was favorable for the adsorption of alkali metals [58]. In a word, combined with XRD, SEM and TEM images of fresh and reduced 1.5Na-Fe1Zn1 catalyst, it can be concluded that the reduction of fresh 1.5Na-Fe1Zn1 catalyst led to the transformation of bulk spinel ZnFe₂O₄ nanoparticles into relatively larger Fe⁰ (260–290 nm) and smaller ZnO (60–70 nm) particles, forming a structure of Fe⁰@ZnO with Na majorly concentrating on ZnO surface. The presence of ZnO partly inhibited the aggregation of Fe⁰ during reduction process, leading to the enhanced dispersion of catalyst particles. Therefore, the enhanced dispersion of Fe⁰ derived from ZnO should be a key reason for the increased stability over Zn-contained iron-based catalysts compared to 1.5Na-Fe catalyst.

3.4.2. Characterizations of spent catalysts after CO₂-FTS reaction

The XRD patterns of the spent 1.5Na-Fe1Zn and 1.5Na0.2S-Fe1Zn

catalyst after 20 h CO₂-FTS reaction are shown in Fig. 10a and b. Under the CO₂-FTS reaction, the Fe⁰ phase was transformed into Fe₃O₄ (PDF#82–1533) and χ -Fe₅C₂ (PDF#89–2544), which were the primary active centers for the RWGS and FTS reactions, respectively, while the ZnO phase remained. By contrast, after the addition of S, the spent 1.5Na0.2S-Fe1Zn catalyst exhibited evident Fe⁰ phase, along with minor Fe₃C (PDF#23–1113) and Fe₅C₂ phase. Notably, Fe₃O₄ phase, the active phase for RWGS reaction, was not detected in the spent S-contained catalyst. To obtain the accurate quantitative analysis of iron oxide and carbide, Mössbauer spectroscopy was employed. As shown in Fig. 10c, d and Table S9, Mössbauer spectroscopy result suggested spent 1.5Na-Fe1Zn1 catalyst consisted of 32 % Fe₃O₄ and 68 % χ -Fe₅C₂, consisting with the XRD results. As for S-contained catalyst, spent 1.5Na0.2S-Fe1Zn1 catalyst consisted of 30.7 % Fe⁰, 28.0 % Fe₃C and 41.3 % χ -Fe₅C₂, which is in line with the XRD results. It suggested that the addition of S promoter inhibited the carburization and oxidation of metallic Fe in CO₂-FTS reaction, leading to the decreased χ -Fe₅C₂ content and the formation of Fe₃C with relative lower carbon content, and no presence of Fe₃O₄. The co-existing of χ -Fe₅C₂ and Fe₃O₄ in 1.5Na-Fe1Zn1 was favorable for CO₂-FTS reaction, contributing to excellent catalytic activity and olefins selectivity. After introducing S promoter, the formed Fe₃C phase exhibited evidently lower FTS activity than Fe₅C₂ [59,60]. Although Fe₃C can serve as active sites for RWGS reaction under high-temperature conditions, its adsorption and activation capability for CO₂ at 320 °C was much lower than that of Fe₃O₄ [8]. Therefore, the absence of Fe₃O₄ phase was also one of the significant reasons for the significant decreased CO₂ conversion. In addition, the presence of S promoter was affecting the CO₂ and CO adsorption and dissociation, which was also responsible for the lower CO₂ conversion and the higher selectivity to CO. Meanwhile, metallic Fe⁰ favored CO₂ hydrogenation into CH₄ and light alkanes [25,61]. Therefore, the presence of a large amount of Fe⁰ would promote alkanes production, leading to the significant decrease in olefins selectivity.

CO₂-TPD and CO-TPD were performed to explore the effect of S promoter on the adsorption of CO₂ and CO. As shown in Figure S6a, two temperature regions from 50 to 250 °C and from 450 to 780 °C for CO₂ desorption could be recognized, corresponding to the weak adsorption and strong chemisorption of CO₂ on catalyst surface, respectively. As for Fe1Zn1 without Na addition, only a weak adsorption peak was observed with smaller peak intensity. After Na addition, the strong high-temperature desorption peak for CO₂ and CO in the range of 450–780 °C appeared, which should be related to strong adsorption of CO₂ and the surface reaction (CO₂*=CO*+O*) on alkali promoted catalytic sites. It suggested that the introduction of Na could efficiently promote the adsorption and dissociation of CO₂ by introducing alkaline sites. When 0.1 wt% Na₂S was added, the peak area for strongly adsorbed CO₂ and the formed CO significantly decreased, indicating that Na₂S suppressed the adsorption and dissociation of CO₂ on the alkali promoted catalytic sites. With the addition of 0.2 wt% Na₂S, the peak position was significantly shifted toward lower temperature, suggesting that S notably weakened the interaction between CO₂/CO and alkaline sites. Similar phenomenon could be observed in CO-TPD (Figure S6b). Fe1Zn1 without Na addition exhibited a small CO desorption peak. After Na addition, this peak largely increased, suggesting the addition of Na also favored CO adsorption on catalyst surface. Apart from CO signal, CO₂ signal could also be recognized in CO-TPD profiles, owing to the dissociation of CO* (2CO*=CO₂*+C*). After adding 0.1 wt% Na₂S, the desorption peak area for CO and CO₂ obviously decreased. When further increasing S content, the peak position was shifted toward lower temperature range. Based on the above results, it indicated that Na promoter enhanced the adsorption and dissociation of both CO₂ and CO over iron-based catalysts, while Na₂S could suppress the adsorption and dissociation of CO₂ and CO. In the FTS reaction, the adsorption and dissociation of CO was highly dependent on the electron back donation from active metallic sites to the antibonding C-O π^* orbitals [33,62]. Such process could be tuned by the addition of promoters with different electronic

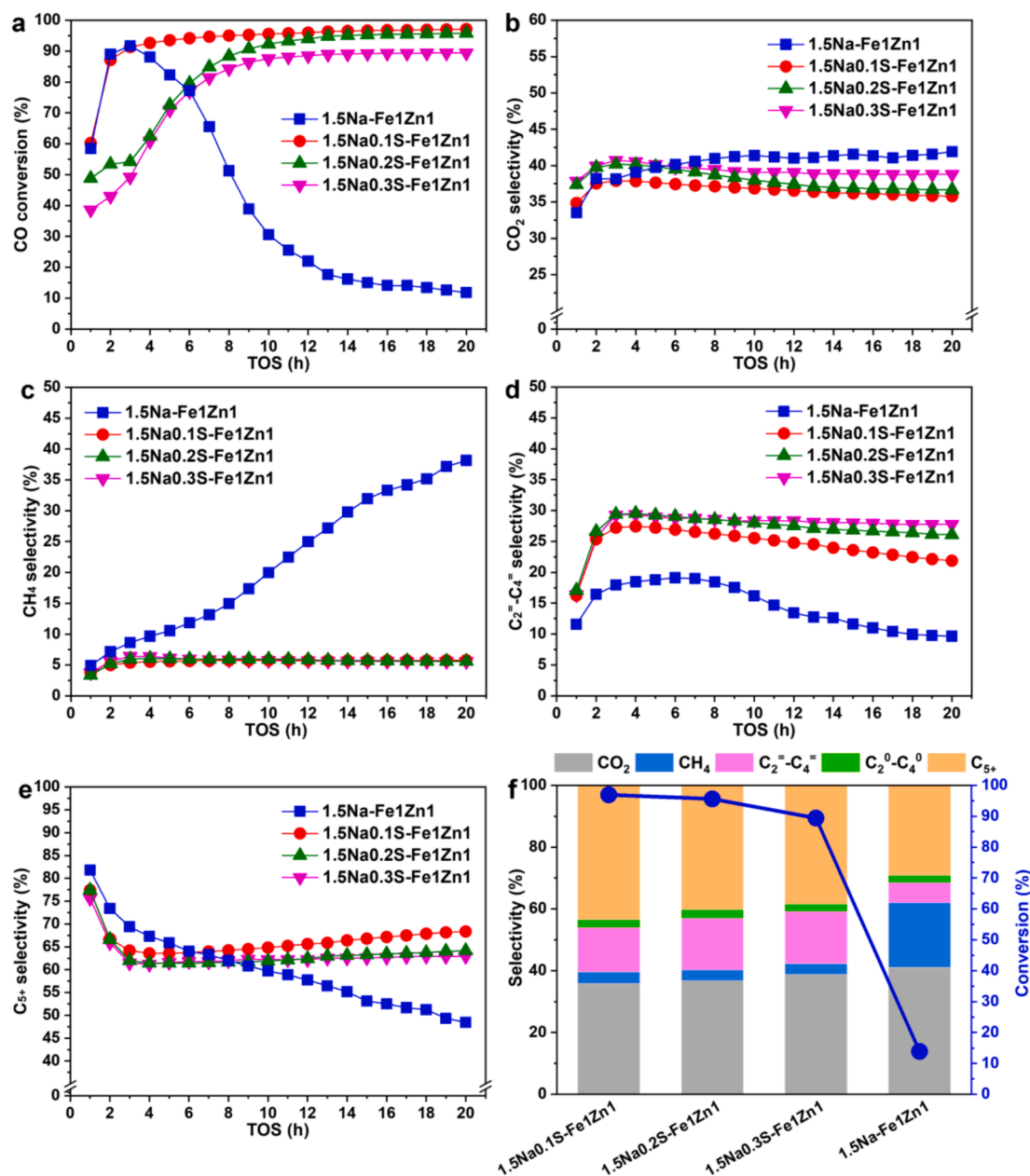


Fig. 11. Catalytic performance of the NaS-Fe1Zn1 catalysts for CO-FTS reaction. Effects of the S content on the (a) catalytic activity, the selectivity of (b) CO₂, (c) CH₄, (d) lower olefins, (e) C₅₊, and (f) product distribution. Reaction conditions: 2.0 g catalyst, 320 °C, H₂/CO = 1, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

properties. The addition of Na promoter into Fe catalyst could increase the electron density of Fe d-orbitals, facilitating its electron back donation to CO₂ and CO antibonding orbitals and contributing to the enhanced adsorption and dissociation of CO₂ and CO. On the contrary, the S promoter exhibited an electron withdrawing effect on Fe catalyst [63]. It would lead to the decrease of the electron density of Fe d-orbitals, which was unfavorable for the adsorption and dissociation of CO₂ and CO. For this reason, the higher S content led to a lower initial activity in CO₂-FTS. Meanwhile, it also has evident effect on the evolution of catalyst phases. According to the previous work, it exists a dynamic structural evolution of iron catalysts involving competitive oxidation and carburization in CO₂-FTS reaction [51]. In the present work, the addition of S evidently suppressed the adsorption and dissociation of CO₂ and CO on catalyst surface, reducing the surface chemical potential of both C and O elements (μ_C and μ_O) and slowing down the subsequent carburization and oxidation of iron phase to give sufficient χ -Fe₅C₂ and Fe₃O₄ for FTS and RWGS reactions, respectively. Correspondingly, more

metallic Fe and Fe₃C phases were generated, which exhibited very limited FTS activity but favored the production of alkanes.

3.5. Study of the NaS-Fe1Zn1 catalysts for CO-FTS reaction

3.5.1. Catalytic performance of the NaS-Fe1Zn1 catalysts for CO-FTS reaction

The NaS-Fe1Zn1 catalysts with different S content were also tested for CO-FTS reaction under similar reaction conditions with CO₂-FTS reaction. As shown in Fig. 11a and Table S10, within the first 3 h of the reaction, the catalytic activity of 1.5Na-Fe1Zn1 without S promoter gradually increased up to its maximum, with a similar trend to that of 1.5Na0.1S-Fe1Zn1. However, 1.5Na0.2S-Fe1Zn1 and 1.5Na0.3S-Fe1Zn1 with higher S content exhibited lower initial activity and longer induction period, indicating the presence of S would slow down the carburization of iron phase. After the induction period, 1.5Na-Fe1Zn1 exhibited evident deactivation, while all the S-contained

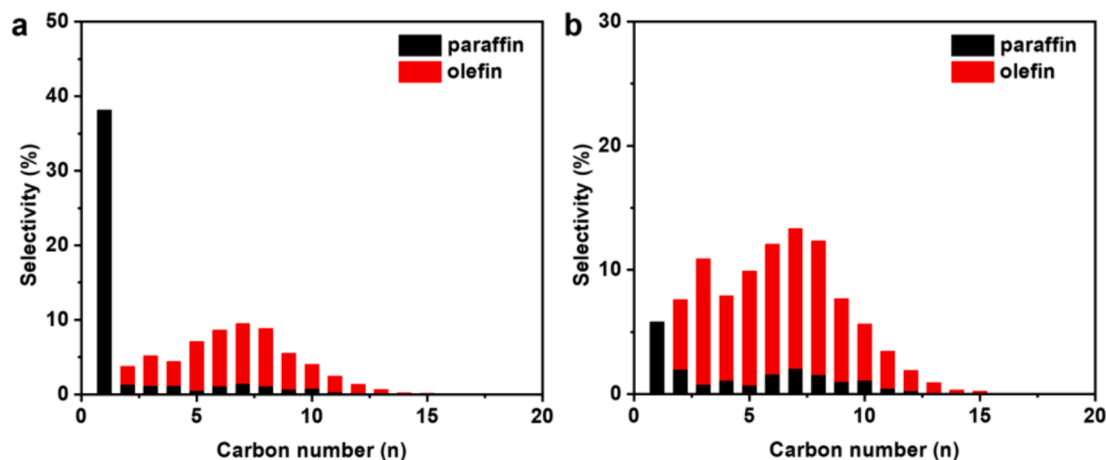


Fig. 12. Carbon number distribution of the total products for CO-FTS reaction over (a) 1.5Na-Fe1Zn1 and (b) 1.5Na0.1S-Fe1Zn1 catalyst. Reaction conditions: 2.0 g catalyst, 320 °C, $H_2/CO = 1$, 1 MPa, and GHSV = $1200 \text{ mL} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$.

catalysts showed excellent catalytic stability. Furthermore, with the increase of S content, catalysts activity gradually decreased.

The product selectivity is presented in Fig. 11b-f. 1.5Na-Fe1Zn1 showed evident increasing trend for the selectivity of CO_2 and CH_4 with catalyst deactivation, while the S-contained catalysts exhibited relatively lower and stable selectivity for CO_2 and CH_4 . Additionally, along with catalyst deactivation, the selectivity of lower olefins gradually decreased on 1.5Na-Fe1Zn1, while they were relatively stable over the S-contained catalysts. With the increase of S content, the selectivity of lower olefins gradually increased, accompanied with the decline in C_{5+} production with unchanged CH_4 selectivity. The carbon distribution of the total products in CO-FTS reaction over 1.5Na-Fe1Zn1 and 1.5Na0.1S-Fe1Zn1 is shown in Fig. 12. Compared to 1.5Na-Fe1Zn1, 1.5Na0.1S-Fe1Zn1 exhibited much higher olefins selectivity along with much lower CH_4 selectivity. Especially, the selectivity to higher-value C_{5+} long-chain olefins was calculated to be 58.6 %. It suggested the crucial role of Na_2S promoter in selective and stable transformation of syngas into higher-value C_{5+} long-chain olefins over the Fe-based CO-FTS reaction [31,64–67].

In conclusion, the catalytic results over NaS-Fe1Zn1 catalysts for CO-FTS reaction were totally different from that for CO_2 -FTS reaction. 1.5Na-Fe1Zn1 catalyst, which exhibited excellent CO_2 -FTS performance, experienced significant deactivation during the CO-FTS reaction. During catalyst deactivation, its chain growth ability decreased significantly, along with the increased selectivity to CO_2 , especially CH_4 . Notably, the selectivity of CH_4 and C_{5+} was more sensitive to catalyst deactivation than that for lower olefins. The addition of S would significantly improve catalyst stability on the CO-FTS reaction. Furthermore, the increase of S content can significantly enhance the selectivity of lower olefins while maintaining CH_4 selectivity. In the case of CO_2 -FTS reaction, the addition of S content significantly decreased catalyst activity, along with the rapid shifting of production selectivity from light olefins to light alkanes and CO.

Furthermore, the effect of reaction conditions on CO-FTS reaction over 1.5Na0.1S-Fe1Zn1 catalyst was screened. As shown in Figure S7 and Table S11, with the increase of reaction temperature from 280 °C to 320 °C, CO conversion increased from 24.0 % to 96.9 %, along with the increase in C_{5+} hydrocarbon selectivity from 63.5 % to 67.8 %. The increased reaction temperature facilitated the dissociation and hydrogenation of CO^* , resulting in improved carburization of iron phase and increased CO conversion. It also resulted in the increase of CH_x^* concentration on catalyst surface, favoring carbon chain growth. As shown in Figure S8 and Table S12, the increasing H_2/CO ratio favored the hydrogenation of surface CH_x^* , leading to the increased CH_4 selectivity and decreased C_{5+} hydrocarbon selectivity. At the same time, the

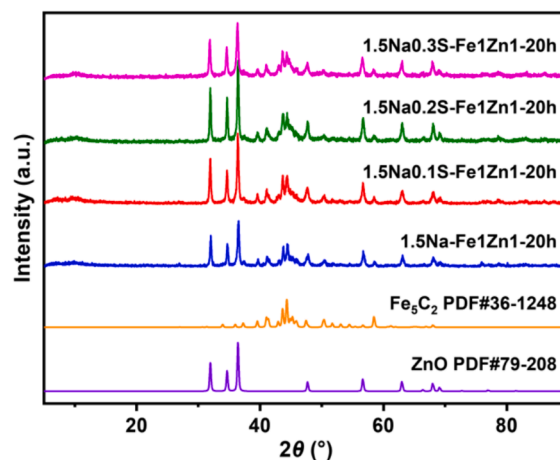


Fig. 13. XRD patterns of the spent NaS-Fe1Zn1 catalysts after CO-FTS reaction.

selectivity to CO_2 was obviously lowered with a higher H_2/CO ratio, which might be owing to the suppressed Boudouard reaction route with a higher surface C^* species concentration for the formation of CO_2 ($CO \rightarrow C^* + O^*$; $O^* + CO^* \rightarrow CO_2$) [39]. In general, compared to H_2/CO_2 feed gas, a higher CO concentration in feed gas, such as synthesis gas, was favorable for the long-chain olefins production, which was ascribed to resulting in surface higher active phase iron carbides concentration and higher polymerization intermediate CH_x^* concentration for chain growth. It can be seen in Figure S9 and Table S13 that a higher reaction pressure could also increase CH_x^* concentration on catalyst surface, leading to the product selectivity shifting to C_{5+} hydrocarbons.

3.5.2. Characterizations of the spent NaS-Fe1Zn1 catalysts for CO-FTS reaction

To explore the effect of Na_2S promoter on CO-FTS reaction, various characterizations were performed including XRD, Mössbauer spectroscopy, TG, O_2 -TPO, Raman spectroscopy, TEM and EDX mapping. XRD results of spent catalysts are shown in Fig. 13, χ - Fe_5C_2 and ZnO phases could be recognized over all the spent catalysts after CO-FTS reaction. Meanwhile, no characteristic diffraction peaks for Fe_3O_4 phase was detected over all the catalysts, indicating much stronger carburization ability in CO-FTS with high CO concentration in feed gas compared to that for CO_2 -FTS. Furthermore, with the increase of Na_2S content, Fe^0 and Fe_3C phases corresponding to the poor catalytic performance in CO_2 -FTS over the Na_2S -contained catalysts were also not detected over the spent Na_2S -contained catalysts in CO-FTS. It indicated that the

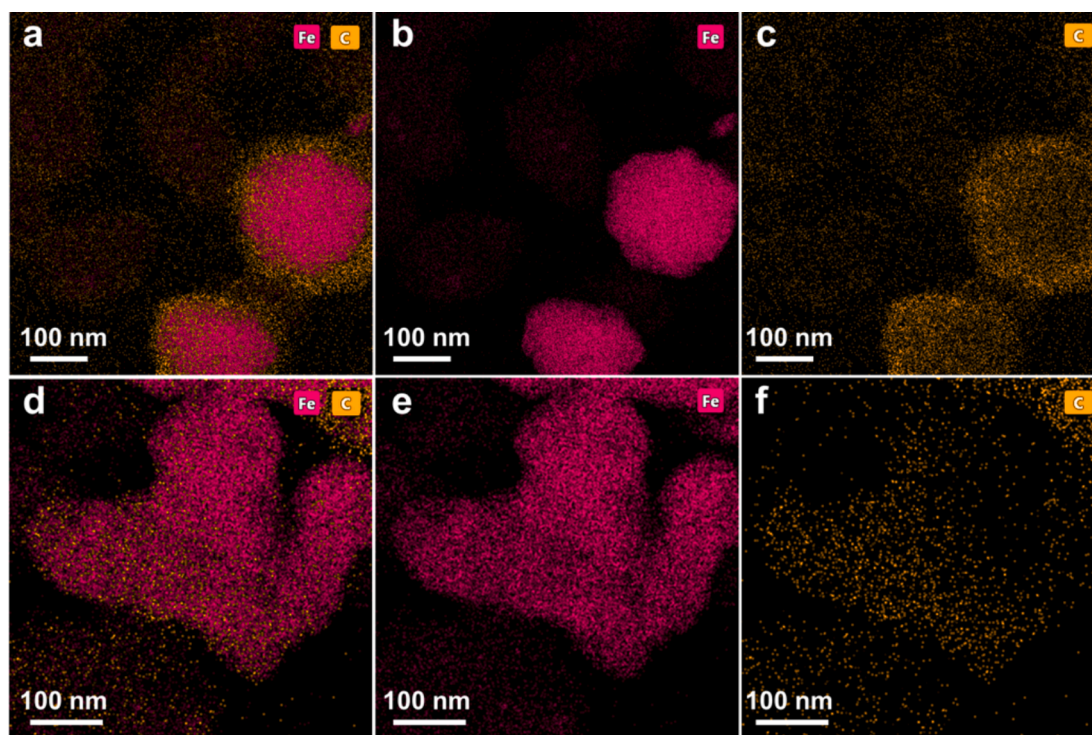


Fig. 14. EDX-mapping images of the spent 1.5Na-Fe1Zn1 (a-c) and 1.5Na0.1S-Fe1Zn1 (d-f) after 20 h CO-FTS reaction.

addition of 0.3 wt% S did not have evident effect on the carburization and resulted iron phase composition over spent NaS-Fe1Zn1 catalysts in spite of that the addition of Na₂S can affect the adsorption and dissociation of CO.

According to the Mössbauer spectroscopy in Figure S10 and Table S14, spent 1.5Na-Fe1Zn1 consisted of 85.5 % χ -Fe₅C₂, 12.1 % Fe₃C and 2.4 % iron oxides, which was similar to that for spent 1.5Na0.1S-Fe1Zn1 with 85.0 % χ -Fe₅C₂, 10.4 % Fe₃C and 4.6 % iron oxides. It further confirmed that the introduction of Na₂S promoter had no significant influence on the phase composition of catalyst for the CO-FTS reaction. Considering the rapid deactivation of 1.5Na-Fe1Zn1 in CO-FTS, it suggested that the transformation of iron phase was not the primary reason for catalyst deactivation.

In fact, carbon deposition is one of the significant reasons for the deactivation of iron-based FTS catalysts [68–70]. The thermal gravimetric (TG) analysis under an O₂ atmosphere was performed over all spent catalysts (Figure S11a). Comparing the four curves, it was evident that the weight of the three S-containing catalysts showed an increasing trend during the heating process, whereas the weight of 1.5Na-Fe1Zn1 started decreasing after the temperature exceeds 250 °C. Two oxidation processes existed in TG testing, including the oxidation of iron carbide species to give iron oxides and the oxidation of carbon species to release carbon oxides. The former was a weight-gaining process, while the latter was a weight-losing process. The weight loss observed in TG analysis of 1.5Na-Fe1Zn1 was related to the release of more carbon oxides during the oxidation of O₂. Based on Mössbauer spectroscopy, the contribution of iron carbide oxidation on TG profiles could be calculated. After excluding this contribution, the calculated weight loss in TG profiles was shown in Figure S11b. Clearly, spent 1.5Na-Fe1Zn1 exhibited much higher weight loss than that of spent 1.5Na0.1S-Fe1Zn1. It suggested that spent 1.5Na-Fe1Zn1 consisted of much more carbon deposition, which was a key reason for catalyst deactivation.

As shown in Figure S12, O₂-TPO curves could be divided into two peaks. The peak at about 300 °C was assigned to the oxidation of carbon deposition, while the peak at 360 °C was related to the oxidation of iron carbide [71]. Clearly, the spent 1.5Na-Fe1Zn1 showed the highest peak

intensity at 300 °C, confirming the existence of large amount of carbon deposition. With the increase of S content, the peak intensity for the oxidation of carbon deposition significantly decreased, suggesting that the addition of S could inhibit carbon deposition. This phenomenon was attributed to the weakened adsorption and dissociation of CO derived from S addition, as evidenced in CO-TPD in Figure S6b. Raman profiles of spent NaS-Fe1Zn1 catalysts are shown in Figure S13. Stronger G band at 1600 cm⁻¹ and D band at 1313 cm⁻¹ could be observed over spent 1.5Na-Fe1Zn1, further confirming the existence of large amount of carbon deposition [72,73].

The TEM images of spent 1.5Na-Fe1Zn1 and 1.5Na0.1S-Fe1Zn1 are presented in Figure S14. It can be observed that the catalyst nanoparticles in spent 1.5Na-Fe1Zn1 were covered with a large amount of carbon species, forming a carbon shell structure with a thickness of up to 28 nm, indicating a significant carbon deposition during the reaction. In contrast, no obvious carbon shell structure was observed in the spent 1.5Na0.1S-Fe1Zn1 catalyst, indicating that the introduction of S can significantly inhibit carbon deposition on the catalyst surface during the CO-FTS reaction. Furthermore, spent 1.5Na-Fe1Zn1 catalyst exhibited relatively smaller particle sizes compared to spent 1.5Na0.1S-Fe1Zn1 catalyst. It can be attributed to the existence of carbon shell, which inhibited particle aggregation and growth during the CO-FTS reaction. This result might indicate that the aggregation of nanoparticles is not a primary reason for catalyst deactivation in Fe-based CO-FTS reaction. EDX mapping was also used to reveal the distribution of Fe and C elements in the spent catalysts. As shown in Fig. 14, it can be observed that Fe element distribution in 1.5Na-Fe1Zn1 was relatively concentrated, forming spherical particles with a diameter of 210 nm. A large amount of C element was distributed on the external surface of these particles, forming a structure of Fe@C. It was consistent with TEM results. By contrast, Fe element distribution in 1.5Na0.1S-Fe1Zn1 was more dispersed, forming larger block structures with C uniformly distributed within Fe. Although carbon shell could restrict the growth of iron particles during the reaction, it also covered the active iron carbide sites, reducing the number of exposed active sites and leading to a substantial decrease in catalytic activity. After the introduction of a trace amount of

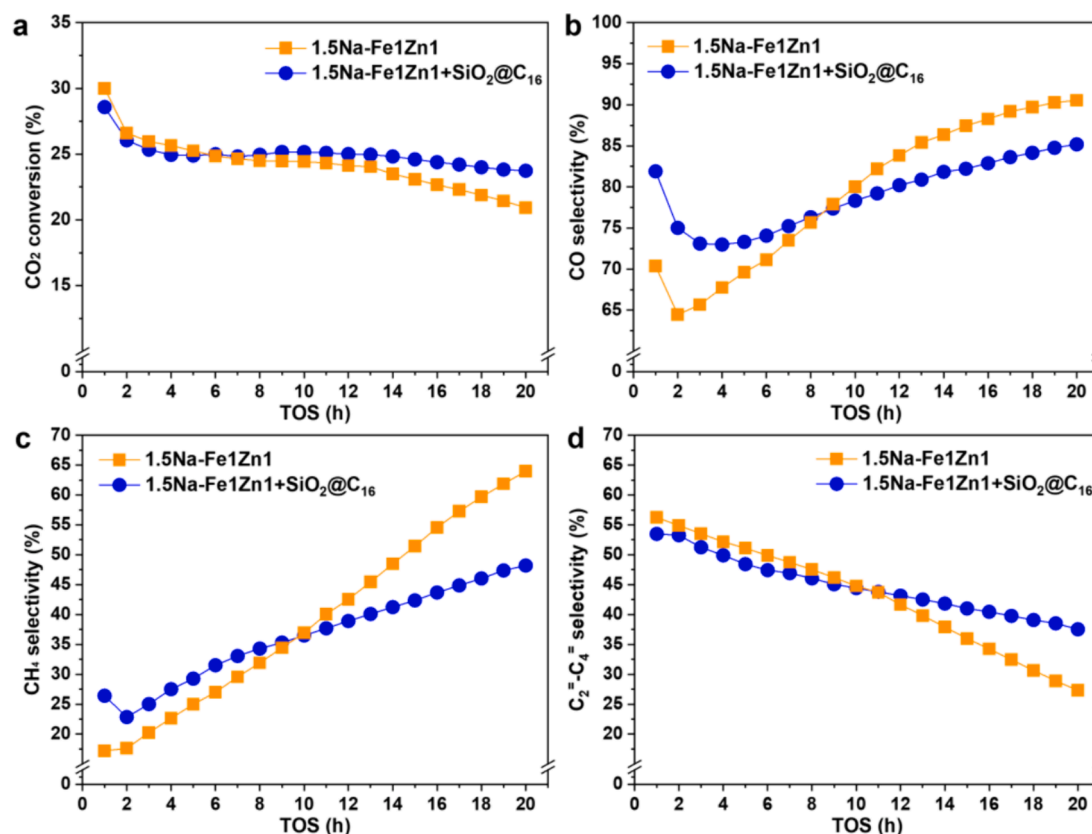


Fig. 15. Catalytic performance of the 1.5Na-Fe1Zn1 and 1.5Na-Fe1Zn1 + SiO₂@C₁₆ catalyst: (a) catalytic activity, and the selectivity of (b) CO, (c) CH₄, and (d) lower olefins. Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 atm, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹.

S, the rate of carbon deposition was significantly reduced, favoring catalyst stability.

As for high-temperature (>280 °C) FTS reaction, carbon deposition generally occurs through the Boudouard reaction ($2\text{CO} \rightarrow \text{C}^* + \text{CO}_2$) [39]. Previous research has shown that the presence of alkali metal promoters (potassium and sodium carbonate) can promote the Boudouard reaction, accelerating the formation of surface carbon [74]. The addition of alkali metal promoters not only facilitates the adsorption and dissociation of CO but also promotes the formation of active iron carbide in the iron-based FTS reaction [75]. It also inhibits the hydrogenation of surface carbon species, leading to the rapid accumulation by coupling of surface free carbon species (C*) on catalyst surface. The resulting dense carbon shell structure on the iron phase surface not only leads to the coverage of active sites, along with catalyst deactivation, but also hinders the C-C coupling of surface hydrocarbon intermediates, reducing the chain growth capability [68,69]. It leads to a decrease in C₅₊ selectivity and an increase in CH₄ selectivity. The addition of S significantly reduces the adsorption amount and strength of CO on the alkaline sites, weakening the promoting effect of Na promoter. It thus inhibits the generation of surface carbon deposition, resulting in the enhanced stability. Moreover, the addition of S has shown promotional effect on the lower olefins production by decreasing the selectivity of C₅₊ without the increase of CH₄, owing to the enhanced termination of chain growth from β -hydride abstraction [31,66].

3.5.3. Comparison of the Na₂S effect in NaS-Fe1Zn1 catalysts between CO₂-FTS reaction and CO-FTS reaction

Comparing the catalytic results of 1.5Na-Fe1Zn1 and S-contained 1.5Na-Fe1Zn1 catalyst in CO₂-FTS and CO-FTS reaction, it concluded that the addition of trace amount of Na₂S significantly reduced catalyst activity in CO₂-FTS reaction but greatly enhanced the stability in CO-FTS reaction. With the increase of Na₂S content, both reactions

showed a decreasing trend in catalytic activity, while more evident decline was observed in CO₂-FTS reaction compared to CO-FTS reaction. This phenomenon was related to different CO concentration in feed gas and resulted phase evolution between the two reaction systems. In the CO₂-FTS reaction, the introduction of trace amount of Na₂S could shift the Fe phase composition from 32.0 % Fe₃O₄ and 68.0 % χ -Fe₅C₂ to 30.7 % Fe⁰, 28 % Fe₃C, and 41.3 % χ -Fe₅C₂, owing to the weakened adsorption and dissociation of CO₂ and CO on catalyst surface and the subsequent decline in the surface μ_{C} and μ_{O} . In contrast, in CO-FTS reaction with much higher CO concentration, Na₂S addition did not show significant effect on the Fe phase composition (85.5 % χ -Fe₅C₂, 12.1 % Fe₃C, and 2.4 % iron oxides for 1.5Na-Fe1Zn1 vs 85.0 % χ -Fe₅C₂, 10.4 % Fe₃C, and 4.6 % iron oxides for 1.5Na0.1S-Fe1Zn1-20 h). It might be related to the much higher μ_{C} along with much lower μ_{O} in CO-FTS. Since competitive carburization and oxidation existed in the structural evolution of iron catalysts, the increase in μ_{C} but decline in μ_{O} would significantly promote carburization process, which consequently minimized the negative effect of Na₂S on iron phase evolution.

On the other hand, without the addition of Na₂S, 1.5Na-Fe1Zn1 did not exhibit significant carbon deposition after 20 h of CO₂-FTS but formed a dense carbon shell after CO-FTS. As shown in the CO-TPD result (Figure S6b), Na-contained 1.5Na-Fe1Zn1 catalyst had a strong ability to adsorb and dissociate CO on the alkali-promoted iron sites. For CO-FTS reaction, the higher CO concentration resulted in an excessive surface free carbon species, leading to the rapid accumulation of surface free carbon species to form dense carbon shell. The addition of Na₂S significantly weakened the adsorption and dissociation of CO, and thus led to a substantial improvement in catalyst stability.

Moreover, the addition of Na₂S has also shown different effect on the product selectivity over NaS-Fe1Zn1 catalysts between CO₂-FTS and CO-FTS reaction. In CO₂-FTS reaction, the introduction of Na₂S evidently shifted product selectivity from olefins to light alkanes and CO, while S

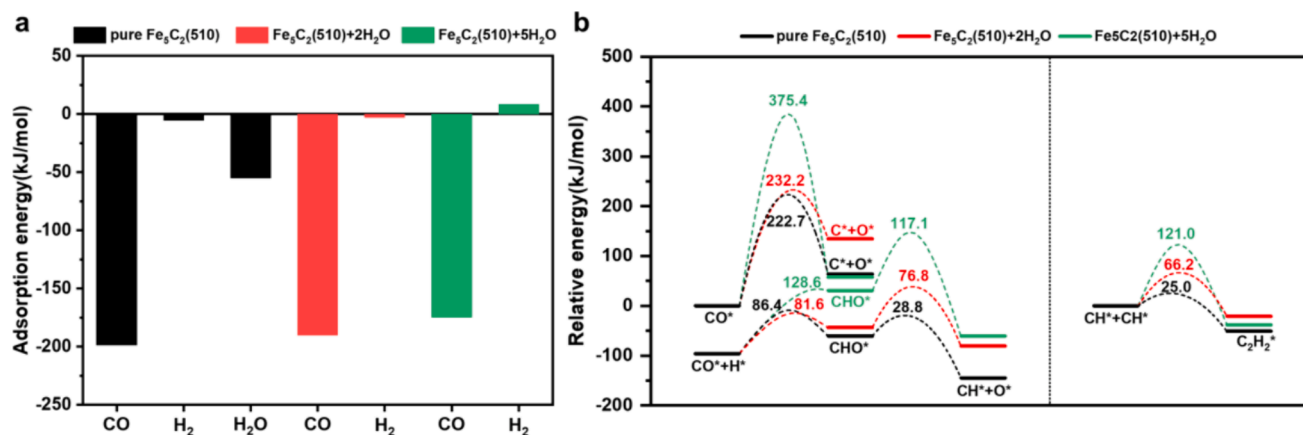


Fig. 16. Energy profiles for CO dissociation and C-C coupling on pure and H₂O-adsorbed Fe₅C₂(510) surfaces.

promoter improved lower olefins selectivity in CO-FTS reaction, along with the decline in C₅₊, unchanged CH₄ selectivity, and O/(O + P). In fact, the resulting product selectivity is a very complicated topic which involves various factors including the electronic effect of Na₂S, the change of catalyst phase evolution derived from Na₂S addition, and the effect of Na₂S on carbon chain growth. In CO-FTS reaction, the effect of Na₂S addition on iron-based catalyst phase evolution could be negligible, considering the similar phase composition between spent 1.5Na-Fe1Zn1 and 1.5Na0.1S-Fe1Zn1. Meanwhile, it could be inferred that the electronic effect of S promoter did not show obvious impact on the product selectivity evidenced by unchanged CH₄ selectivity and O/(O + P), although it was considered that the electron-donating suppression of Na onto iron active sites by the addition of Na₂S might promote the hydrogenation to give more alkanes. Therefore, the enhanced lower olefins selectivity is reasonably attributed to the β -hydride abstraction promoted chain termination. In CO₂-FTS reaction, since the electronic effect of Na₂S on catalyst hydrogenation ability could be ignored, the shifting of product selectivity from olefins to light alkanes is confirmed to be closely related to the change of catalyst iron phases derived from the decline in surface μ_C and μ_O through Na₂S addition, which consequently resulted in the presence of metallic Fe phase with stronger hydrogenation ability for light alkanes production. In summary, based on the rationally designed experiments, the effect of Na₂S element on product selectivity in CO₂-FTS and CO-FTS reactions has been clearly elucidated.

3.6. Effect of the hydrophobic modification on CO₂-FTS reaction

3.6.1. Catalytic results of the hydrophobically modified 1.5Na-Fe1Zn1 catalyst for Atmospheric-Pressure and High-Pressure CO₂-FTS reactions

In comparison to CO-FTS, CO₂-FTS reaction generally produces about two times of water from RWGS and CO-FTS reaction. Such large amount of water has significant effect on the carburization and oxidation of iron phase and the catalytic performance. In order to weaken the negative effect of water, we proposed to develop a new strategy of the hydrophobic modification by simply mixing the FTS catalyst with hydrophobic-treated diluent SiO₂ to investigate the effect of water on CO₂-FTS reaction. As shown in Figure S15, the hydrophobic treatment of SiO₂ was achieved through the silanization reaction with hexadecyltrimethoxysilane reported in our previous work [5]. As shown in Figure S16, after the introduction of hydrophobic layer, the contact angle with water increased from 27.5° to 125.5°, suggesting the silanization treatment successfully shifted the surface of SiO₂ from hydrophilicity to hydrophobicity.

As shown in Fig. 15, after the hydrophobic modification, the 1.5Na-Fe1Zn1 + SiO₂@C₁₆ exhibited an improved stability under atmospheric pressure compared to 1.5Na-Fe1Zn1. Under the atmospheric-pressure

reaction condition, it is thermodynamically unfavorable for the CO hydrogenation reaction into olefins over Fe-based catalysts, leading to a large amount of non-dissociated CO on catalyst surface. For the 1.5Na-Fe1Zn1 with hydrophilic surface, the formed H₂O tended to accumulate on catalyst surface, leading to a higher surface H₂O concentration. The high concentration of CO and H₂O would hinder subsequent RWGS reaction both thermodynamically and dynamically, which contributed to the gradual catalyst deactivation. After the hydrophobic modification with SiO₂@C₁₆, the introduced hydrophobic surface exhibited much weaker interaction with H₂O, resulting in the increased mass transfer rate of byproduct H₂O to leave FTS catalyst surface [34,76]. It could weaken the negative effect of water on atmospheric-pressure CO₂ hydrogenation via promotional RWGS reaction and thus enhance catalyst stability. The enhanced mass transfer of H₂O through hydrophobic-modified diffusion channel can be demonstrated by the dehydration experiments of CuSO₄·5H₂O and the FTIR spectra of spent catalysts (Figures S17 and S18). As for selectivity, both the catalysts exhibited a decline in CO selectivity at the first 2 h of reaction and gradually increasing trend from 3 h to 20 h. The decreasing CO selectivity in the initial stage of reaction was related to the carburization of iron phase with the consumption of an extra amount of CO, while the increasing CO selectivity after the induction period was attributed to the inhibitory hydrogenation and dissociation of CO* species at atmospheric-pressure reaction condition. Indeed, the low surface H₂ concentration at atmospheric pressure was unfavorable for the C* hydrogenation and subsequent C-C coupling. Comparing 1.5Na-Fe1Zn1 and 1.5Na-Fe1Zn1 + SiO₂@C₁₆, 1.5Na-Fe1Zn1 without hydrophobic modification exhibited a more rapid increasing trend of CO selectivity, which suggested that the presence of catalyst surface water suppressed the adsorption and hydrogenation of CO. Additionally, both the catalysts exhibited an increasing CH₄ selectivity and decreasing lower olefins selectivity after the induction period, suggesting the decreasing chain growth ability with time on stream. Such phenomenon was also observed in the reported ambient-pressure CO₂-FTS reaction [8]. It was attributed to the accumulation of inactive carbon species on catalyst surface, which would inhibit the possibility of C-C coupling. After the hydrophobic modification, the fast desorption of water from catalyst surface would strengthen the hydrogenation of dissociative C* species to form more CH_x* and benefit for subsequent C-C coupling, leading to the evidently slower increase in CH₄ selectivity and slower decrease in lower olefins selectivity over 1.5Na-Fe1Zn1 + SiO₂@C₁₆.

However, in the case of high-pressure, the hydrophobic modification of catalysts did not show promotional effect on CO₂-FTS reaction over 1.5Na-Fe1Zn1 + SiO₂@C₁₆. As shown in Fig. 4, Figure S19, and Table S15, both 1.5Na-Fe1Zn1 and 1.5Na-Fe1Zn1 + SiO₂@C₁₆ indicated stable CO₂ conversion and product selectivity. At high reaction pressure, the dissociated C* species from CO intermediate could be much easier to

hydrogenate to form a large amount of surface CH_x^* intermediates for subsequent C–C coupling. Therefore, the inactive C^* species accumulation was obviously suppressed to achieve the stability of CO_2 conversion and product selectivity. Reasonably, the presence of water would indicate minor impact on the catalytic performance in the high pressure CO_2 -FTS reaction.

3.6.2. SSW-NN simulation on H_2O effect

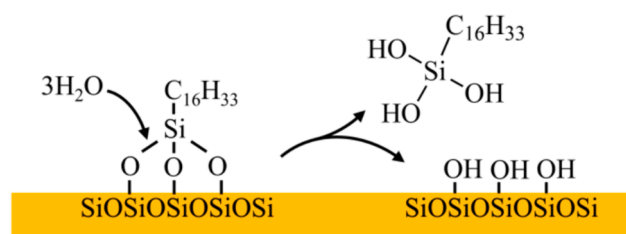
To obtain a molecule-level understanding of the effect of H_2O on FTS reaction, SSW-NN simulation was conducted. Compared to DFT calculations, this simulation method has shown much higher efficiency in potential energy surface (PES) exploration with a comparable accuracy, which has been successfully applied in the theoretical study of Fe-based FTS reaction [48]. In the present work, three slab models, including pure $\text{Fe}_5\text{C}_2(510)$ surface, $\text{Fe}_5\text{C}_2(510)$ surface adsorbed with two H_2O molecules, and $\text{Fe}_5\text{C}_2(510)$ surface adsorbed with five H_2O molecules were constructed in order to elucidate the effect of H_2O . As shown in Fig. 16a and Figure S20, the adsorption energy of molecular CO , H_2 , and H_2O on pure $\text{Fe}_5\text{C}_2(510)$ surface was calculated to be -198.7 kJ/mol, -5.5 kJ/mol, and -55.1 kJ/mol, respectively. With the increase of adsorbed H_2O , it was found that adsorption energies of CO and H_2 increased, suggesting that the accumulation of H_2O molecules on catalyst surface would weaken the interaction between CO/H_2 and catalyst surface. Meanwhile, as shown in Figure S21, with the increase of adsorbed H_2O , the dissociation energy of H_2 exhibited an increasing trend. It suggested that H_2O accumulation also suppressed H_2 dissociation, which was unfavorable for subsequent hydrogenation reaction in FTS reaction.

As shown in Fig. 16b and Figures S22–24, the effects of H_2O accumulation on CO dissociation and hydrogenation and C–C coupling of CH^* were also considered. It was found that with the increase of adsorbed H_2O molecules on $\text{Fe}_5\text{C}_2(510)$ surface, the energy barriers for CO direct dissociation ($\text{CO}^* \rightarrow \text{C}^* + \text{O}^*$) and H-assisted dissociation ($\text{CO}^* + \text{H}^* \rightarrow \text{CHO}^*$; $\text{CHO}^* \rightarrow \text{CH}^* + \text{O}^*$) exhibited an increasing trend. It suggested that water accumulation was also unfavorable for the dissociation of CO . It can be inferred that in the atmospheric-pressure CO_2 -FTS reaction, the gradual accumulation of H_2O on catalyst surface suppressed the adsorption and dissociation of CO and H_2 , leading to the gradual catalyst deactivation along with higher CO selectivity. Moreover, with the increase of adsorbed H_2O molecules, the energy barriers for CH^* coupling increased from 25.0 kJ/mol to 121 kJ/mol. It confirmed the negative effect of H_2O accumulation on chain growth process, which was responsible for the continuously and linearly increased selectivity towards CH_4 along with catalyst deactivation.

Owing to the presence of H_2O , more CO^* and CHO^* species will exist in catalyst surface since H_2O molecules can suppress H-assisted CO dissociation. Therefore, their effect on the CH^* coupling was also considered. As shown in Figures S25–27, the energy barriers for CH^* coupling on CO^* and CHO^* adsorbed $\text{Fe}_5\text{C}_2(510)$ surfaces were calculated to be 47.0 kJ/mol and 41.3 kJ/mol, respectively, which were higher than that (25.0 kJ/mol) for pure $\text{Fe}_5\text{C}_2(510)$ surface. It further confirmed the negative effect of H_2O on C–C coupling.

3.6.3. Characterization of spent 1.5Na-Fe1Zn1 catalysts

To check the different effect of hydrophobic modification between atmospheric pressure and high pressure conditions, XRD and water contact angle test were performed for the spent catalysts. As shown in Figure S28a, the spent catalysts under atmospheric pressure exhibited similar XRD patterns, majorly consisting of ZnO and Fe_5C_2 phases. It suggests the hydrophobic modification did not have significant effect on the phase composition of catalysts. Similarly in Figure S28b, in the case of high pressure reaction, the spent catalysts also exhibited similar XRD patterns. As shown in Figure S29, after 20 h CO_2 -FTS reaction under atmospheric pressure, the water contact angle of hydrophobic-treated SiO_2 was slightly reduced from 125.5° to 116.8° , confirming its hydrophobicity during the reaction. However, after the reaction under high pressure, its water contact angle was evidently reduced to 69.8° ,



Scheme 1. Illustration of the hydrolysis of hexadecyl groups on $\text{SiO}_2@\text{C}_{16}$ during high-pressure CO_2 -FTS reaction.

indicating that the hydrophobic layer was unstable under high pressure CO_2 -FTS conditions. At high reaction pressure and high reaction temperature, the relatively high CO_2 conversion produced high steam concentration around the hydrophobic layer of SiO_2 , which resulted in the destruction of the hydrophobic layer via the hydrolysis of hexadecyl group as shown in Scheme 1.

3.7. Long-term CO_2 -FTS testing over 1.5Na-Fe1Zn1 catalyst

Finally, a 1000 h CO_2 -FTS testing was performed over 1.5Na-Fe1Zn1 catalyst at 320°C and 1 MPa with a GHSV of $1200 \text{ mL} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ to check out its long-term stability. As shown in Fig. 17a–c, the 1.5Na-Fe1Zn1 catalyst exhibited no significant deactivation for CO_2 conversion during the 1000 h long-term CO_2 -FTS reaction, demonstrating its excellent stability. Interestingly, with the increase of reaction time, selectivity to CO slightly increased from 13.5% to 15.0% . Similarly, selectivity to CH_4 also slightly increased from 13.5% to 19.9% , along with the increase of lower olefins selectivity from 36.1% to 37.7% and the decrease of C_{5+} selectivity from 45.0% to 36.2% . Correspondingly, the collected liquid-phase product was shifting to lower carbon number in C_{5+} and the alkane content in C_{5+} was becoming higher with reaction time on stream from 20 h to 1000 h (Figure S30). As shown in Figure S31 and Fig. 17d, the carbon number distribution of the total products collected in the 20 h and 1000 h run was calculated to follow the Anderson-Schulz-Flory (ASF) law. It is clear that the product selectivity was mainly concentrated in lower olefins with C_2^- – C_5^- of about 45.5% .

To further increase the lower olefins selectivity, a catalytic cracking reaction of the collected liquid-phase CO_2 -FTS products during 1000 h was conducted in the fixed-bed reactor over ZSM-5 nanosheet catalyst with Si/Al ratio of about 200. The XRD patterns and SEM image are shown in Figure S32. As shown in Fig. 17e, the cracking products were mainly concentrated in lower olefins with C_2^- – C_5^- content of about 80.9% and consisted of almost no alkanes owing to the absence of H_2 . The results in Figure S33 indicated that the catalytic cracking reaction was stable over nanosheet ZSM-5 catalyst. It was confirmed that the carbon number distribution of the cracking products was almost unchanged with reaction time on stream and very few coke was deposited in catalyst. Combining the result of CO_2 -FTS reaction with that of catalytic cracking reaction, the carbon number distribution of total products was calculated and shown in Fig. 17f. The selectivity of C_2^- – C_5^- lower olefins reached up to about 64.6% , along with a high CO_2 conversion of 38.7% , and a low CO selectivity of 14.5% .

Presently, there have two routes for selective lower olefins production from CO_2 hydrogenation including the modified-FTS route and methanol-mediated route. As summarized in Table S16, the tandem catalysts with FTS catalysts and zeolites usually exhibit higher CO_2 conversion, possibly due to the shifting of equilibrium in CO_2 -FTS reaction derived from the facile diffusion and subsequent transformation of FTS products over acidic zeolites. However, the selectivity towards lower olefins production is restricted, owing to a weak cracking ability of zeolites at relative lower temperature ($\sim 320^\circ\text{C}$). In spite of that a stronger acidic zeolite catalyst and/or higher reaction temperature could be applied for enhanced conversion of the FTS products, more aromatics and alkanes will be produced owing to the easier

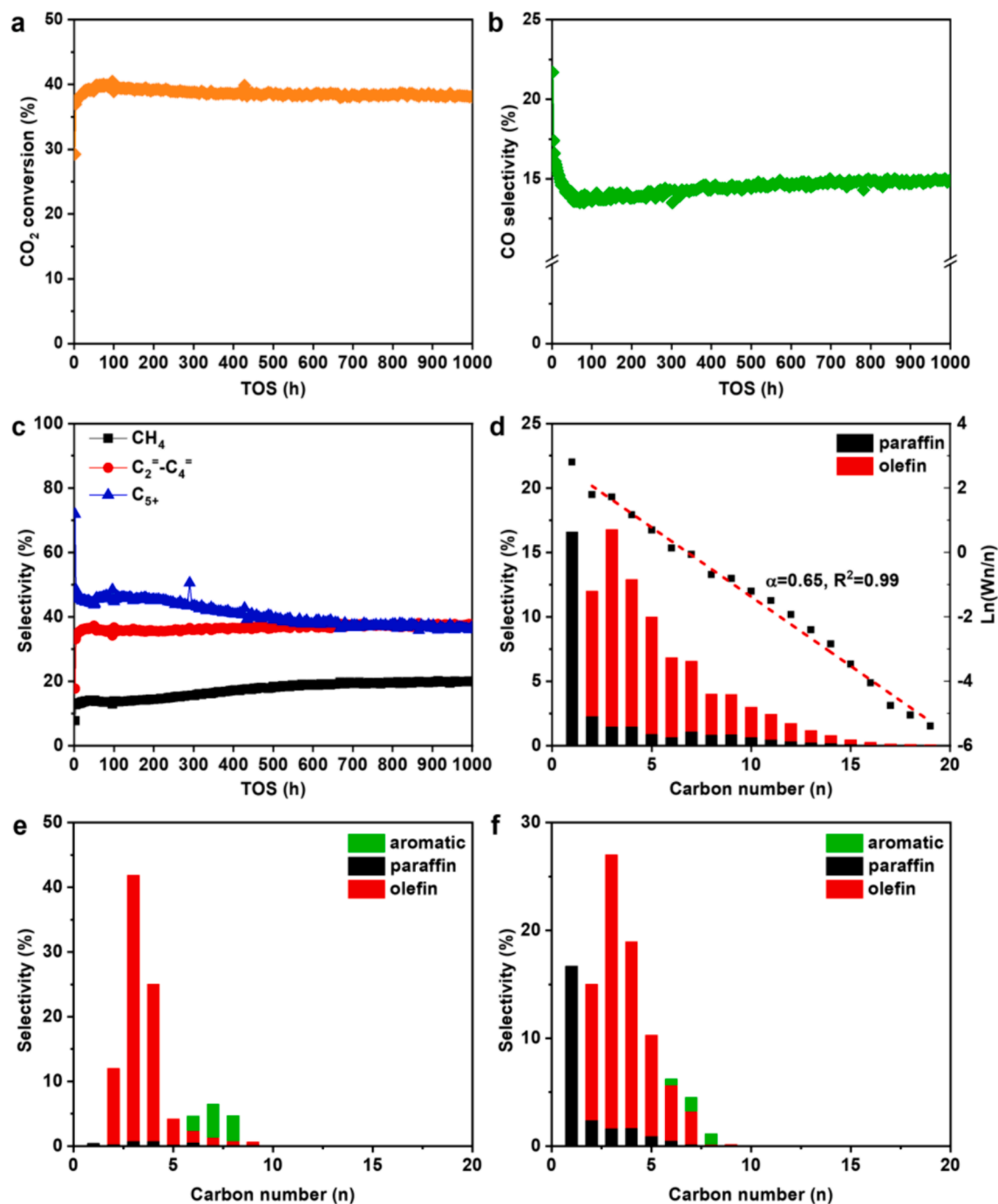


Fig. 17. Catalyst stability testing on 1000 h CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst and the catalytic cracking reaction of collected liquid products: (a) CO₂ conversion, (b) CO selectivity, (c) hydrocarbon product selectivity, (d) carbon number distribution of total products, (e) carbon number distribution of cracking products, and (f) carbon number distribution of total products combining CO₂-FTS reaction with catalytic cracking reaction. Reaction conditions for CO₂-FTS and cracking reaction: 2 g FTS catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL·g_{cat}⁻¹·h⁻¹; 0.15 g zeolite catalyst, 550 °C, and WHSV = 3/h.

dehydrogenation-aromatization and the facile hydrogenation reaction with H₂ environment. Therefore, the suitable catalyst and reaction system for the efficient production of lower olefins should follow the below strategies. On the one hand, it is considered that the designed FTS catalyst gives light olefins formation as high as possible and avoids the excessive formation of undesired CH₄; on the other hand, the collected liquid products are cracked in a separate fixed-bed reactor with a well-designed zeolite catalyst and without H₂ environment. As shown in Table S16, the methanol-mediated route for the production of light olefins from CO₂ hydrogenation shows desired high selectivity. However, the most disadvantages lie in the low CO₂ conversion level and

much higher selectivity towards CO owing to the weak hydrogenation ability with oxides catalyst, and the limitation of thermodynamic equilibrium.

To take insights into crucial reason for the changing product selectivity with time on stream during the long-term test, various characterizations were conducted. Considering a large amount of water generated in CO₂-FTS reaction, it might lead to the loss of alkali metal promoter on catalyst surface. ICP-OES analysis was applied to determine the elemental composition of two spent catalysts after 20 h and 1000 h CO₂-FTS reaction (denoted as 1.5Na-Fe1Zn1-20 h and 1.5Na-Fe1Zn1-1000 h, respectively). As shown in Table S17, 1.5Na-Fe1Zn1-20 h and

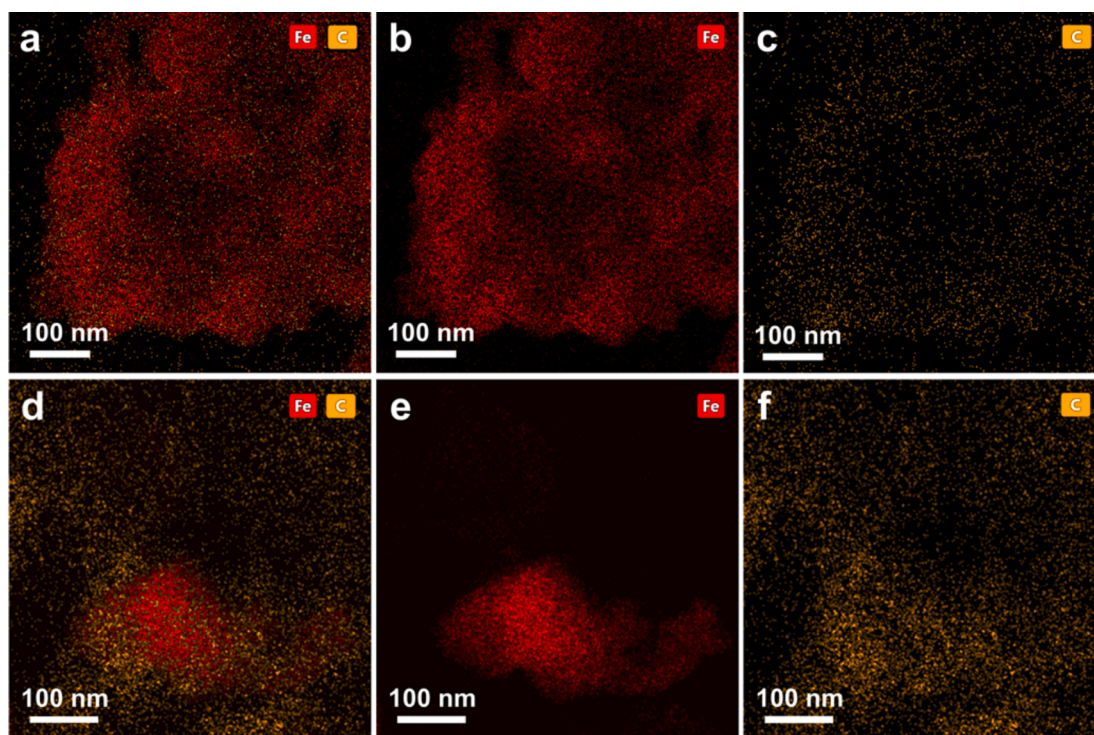
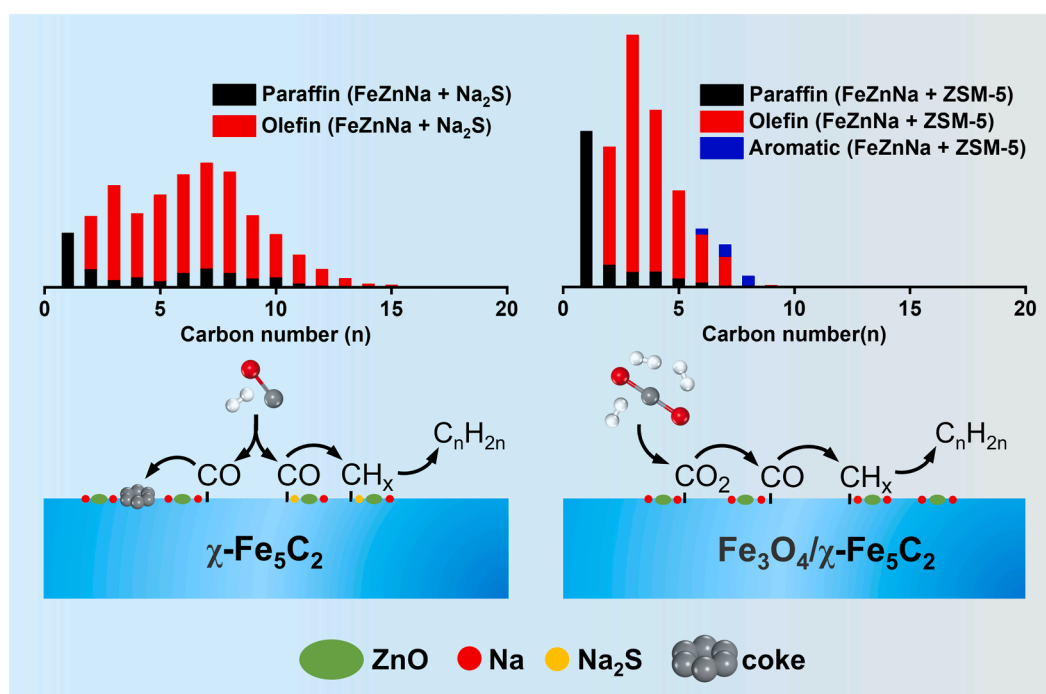


Fig. 18. EDX-mapping images of (a-c) spent 1.5Na-Fe1Zn1-20 h and (d-f) spent 1.5Na-Fe1Zn1-1000 h.

1.5Na-Fe1Zn1-100 h contained 2.1 wt% and 1.8 wt% Na, respectively, which suggested that the loss of Na promoter was not the major reason for changing reaction performance. Iron phase composition was obtained from XRD patterns and Mössbauer spectrums in order to investigate its effect on reaction results. As shown in Figure S34, both the spent catalysts exhibited similar Fe_3O_4 (82–1533), Fe_5C_2 (PDF#89–2544), and ZnO (PDF#79–208) phase. Figure S35 and

Table S18 displayed the result of Mössbauer spectrums. 1.5Na-Fe1Zn1-1000 h consisted of 30 % Fe_3O_4 and 70 % $\chi\text{-Fe}_5\text{C}_2$, which was similar to that for 1.5Na-Fe1Zn1-20 h (32 % Fe_3O_4 and 68 % $\chi\text{-Fe}_5\text{C}_2$). It indicated that 1.5Na-Fe1Zn1 catalyst has excellent iron phase stability for stable CO_2 conversion rate during the long-term CO_2 -FTS testing. We further examined the coke content between the two spent catalysts. TG profiles are presented in Figure S36. 1.5Na-Fe1Zn1-1000 h exhibited much more



Scheme 2. Illustration of the production of long-chain olefins from H_2/CO over Na_2S -modified FeZnNa catalyst and the production of light olefins from H_2/CO_2 via FeZnNa-catalyzed CO_2 -FTS reaction combined with separate catalytic cracking of liquid-phase products over ZSM-5 nanosheet catalyst.

weight loss compared with that of 1.5Na-Fe1Zn1-20 h, suggesting more carbon accumulation on catalyst surface during the long-term FTS reaction. The more carbon accumulation was also confirmed by the O₂-TPO profiles and the Raman spectra as shown in Figures S37 and S38. In addition, in contrast to 1.5Na-Fe1Zn1-20 h, 1.5Na-Fe1Zn1-1000 h exhibited evident carbon accumulation on the external surface of iron nanoparticles (Figure S39). Meanwhile, in the EDX mapping images of the two spent catalysts as shown in Fig. 18, the distribution of Fe and C element in 1.5Na-Fe1Zn1-20 h was very similar and uniformly dispersed. However, in 1.5Na-Fe1Zn1-1000 h, the distribution of Fe element was relatively centralized, while C element was majorly located outside the Fe regions, confirming the existence of carbon-coated Fe structure. It also indicated that the carbon accumulation facilitated the re-dispersion of iron nanoparticles by its filling into the pores of the aggregated iron nanoparticles and finally forming the carbon-coated Fe structure.

Based on the above characterization results, it suggested that the excellent stability for CO₂ conversion during the long-term testing over 1.5Na-Fe1Zn1 catalyst was attributed to stable iron phase composition. The stable Fe₃O₄ and γ -Fe₅C₂ phase derived from the equilibrium between oxidation and carburization during CO₂-FTS reaction facilitated efficient CO₂ hydrogenation into CO via RWGS reaction and subsequent CO-FTS reaction to produce hydrocarbon products. Even so, the product selectivity was still slowly changed during the long-term test. It was considered that the gradually increased carbon accumulation on catalyst surface was a major reason for the shifting product selectivity. Seen from the Raman spectra results of spent catalysts in Figure S38, the much larger D band compared to G band in spent 1.5Na-Fe1Zn1-1000 h catalyst suggested that these deposited carbon species consisted of a large amount of defective carbon sites. These defective sites favored the storage of H* species and thus higher surface H* species concentration on catalyst surface, which resulted in the increasing selectivity to CH₄ and alkanes content in C₅₊ product. In our previous work for CO-FTS reaction over carbon nanotube supported cobalt catalysts, the promotional effect of defective carbon sites on catalytic activity and CH₄ selectivity was also observed, which was well demonstrated due to enhanced hydrogen spillover [77]. The decline in C₅₊ selectivity and shifting to lower carbon number might be also ascribed to the increased carbon deposition, which was accumulated on active iron carbide phase surface to suppress the chain growth.

4. Conclusions

In conclusion, this study has taken comprehensive and systematic insights into the effects of Na₂S and H₂O on the selective and stable production of α -olefins from CO₂/H₂ and CO/H₂ over iron-based catalysts. The key viewpoints have been reached as shown below:

(1) The addition of Na₂S into 1.5Na-Fe1Zn1 catalyst significantly lowered catalyst activity along with light alkanes and CO as major products in CO₂-FTS reaction. It was attributed to the suppression of carburization and oxidation of iron phases with Na₂S addition, leading to lower γ -Fe₅C₂ content, high metallic Fe content with strong hydrogenation ability, and no Fe₃O₄ for RWGS reaction. However, in CO-FTS reaction, Na₂S didn't show obvious impact on the iron phases evolution but significantly improved catalyst stability by suppressing carbon deposition, which was favorable for the selective and stable production of high-value long-chain α -olefins from CO hydrogenation as shown in Scheme 2.

(2) Combined with catalytic tests and theoretical calculations, the effects of H₂O on CO₂-FTS reaction have been clarified as suppressing the adsorption and dissociation of H₂/CO, subsequent hydrogenation to form CH_x*, and their carbon-carbon coupling. Thus, it led to gradual catalyst deactivation in atmospheric-pressure CO₂-FTS reaction, along with continuous and linear increase in CO and CH₄ selectivity and decrease in lower olefins selectivity. Through hydrophobic modification with SiO₂@C₁₆, the favorable desorption of H₂O from catalyst surface

resulted in more stable CO₂ conversion and product selectivity. However, it should be especially pointed out that, in high-pressure CO₂-FTS reaction, the hydrophobic structure of SiO₂@C₁₆ was seriously destroyed owing to the facile hydrolysis of Si-O-Si-C₁₆ with high H₂O steam concentration. Therefore, it is highly expected to develop more stable hydrophobic material with exceptional hydrothermal stability to minimize the negative effects of H₂O in high-pressure CO₂-FTS reaction for viable commercial application.

(3) The optimal 1.5Na-Fe1Zn1 catalyst exhibited excellent stability for reaction activity, along with a slight change of product selectivity during 1000 h long-term CO₂-FTS reaction. The stable iron phase stemmed from the equilibrium between carburization and oxidation contributed to the excellent stability for CO₂ conversion. The increasing carbon deposition for the storage of H* species was the major reason for increasing CH₄ selectivity and alkanes content in C₅₊ and decreasing C₅₊ selectivity, owing to enhancing H* species supply and decreasing C-C coupling. Moreover, in order to further increase lower olefins content, the collected liquid-phase CO₂-FTS products were cracked in a separate fixed-bed reactor over ZSM-5 nanosheet in the absence of H₂. The cracking products were dominantly shifting toward C₂-C₅ lower olefins with minor C₆-C₈ aromatics formation as shown in Scheme 2. Ideally, the designed cracking process almost did not result in the formation of undesired CH₄ and other light alkanes.

CRediT authorship contribution statement

Yufeng Li: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Jiayang Chen:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Zhangshi Li:** Data curation. **Jie Chen:** Data curation. **Yuxin Wang:** Formal analysis, Data curation. **Yuebing Xu:** Data curation. **Bing Liu:** Formal analysis, Data curation. **Yang Jiao:** Project administration. **Xiaohao Liu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcat.2024.115587>.

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